Signals and Systems

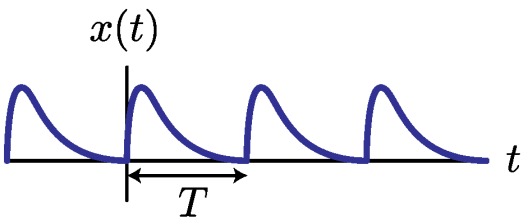
Lecture 12: Continuous-time Fourier Transform

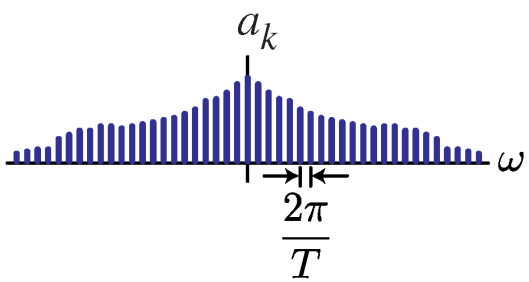
Instructor: Prof. Ting Zhang
Zhejiang University

04/24/2025
Adapted from the materials provided on
the MIT OpenCourseWare

Review: Fourier Series

This week: generalize for aperiodic signals.



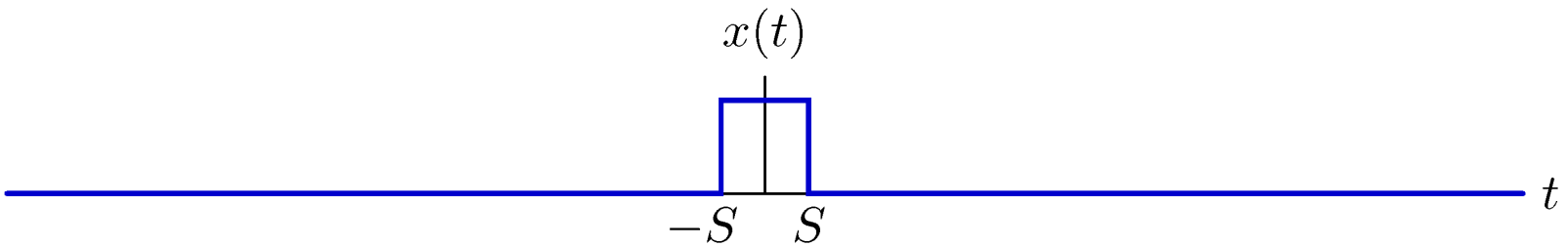


How can we represent an aperiodic signal as a sum of sinusoids?

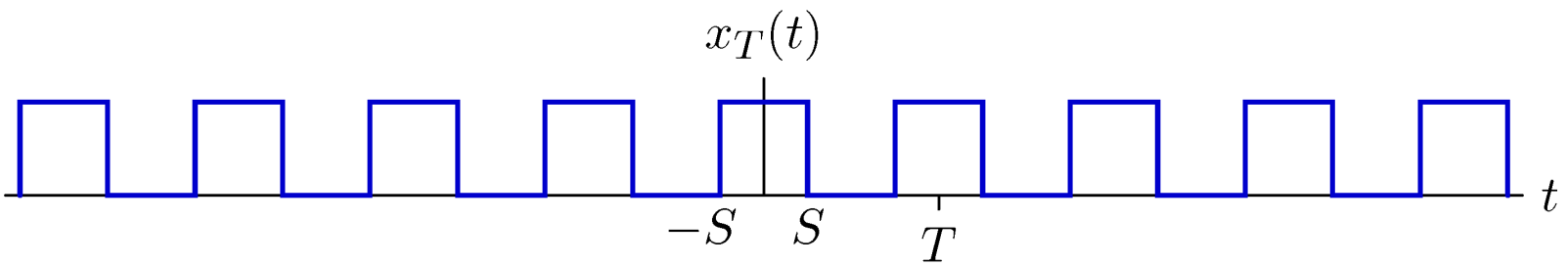
Fourier Transform

An aperiodic signal can be thought of as periodic with infinite period.

Let $x(t)$ represent an aperiodic signal.



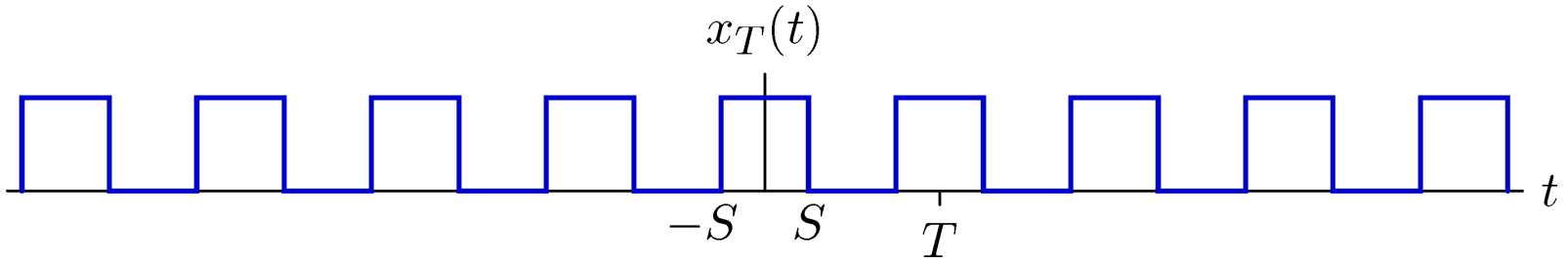
"Periodic extension": $x_T(t) = \sum_{k=-\infty}^{\infty} x(t + kT)$



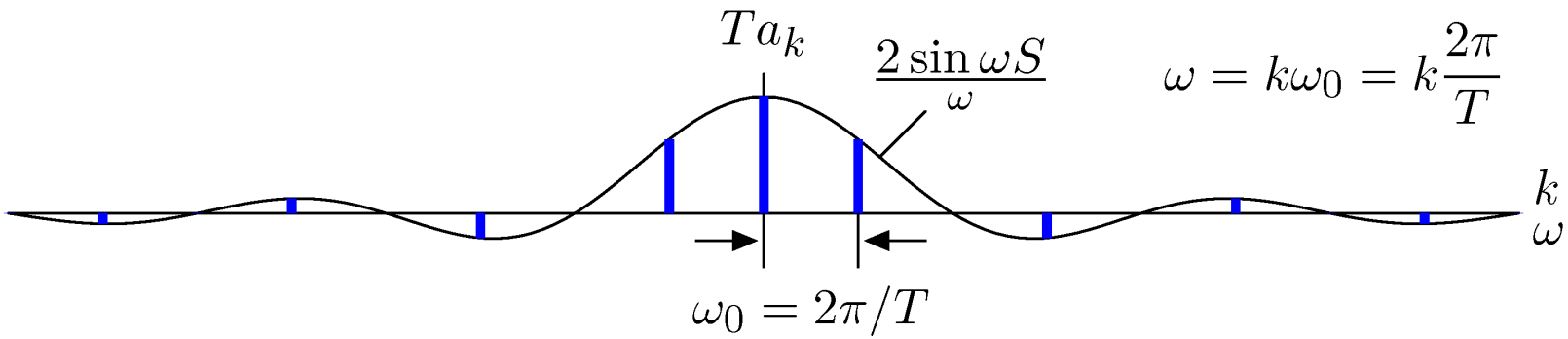
Then $x(t) = \lim_{T \to \infty} x_T(t)$.

Fourier Transform

Represent $x_T(t)$ by its Fourier series.

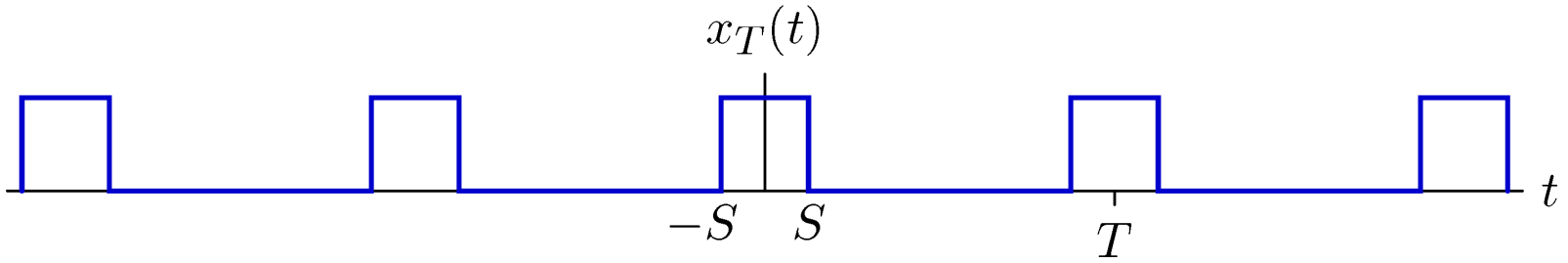


$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} x_T(t)e^{-j\frac{2\pi}{T}kt}dt = \frac{1}{T}\int_{-S}^{S} e^{-j\frac{2\pi}{T}kt}dt = \frac{\sin\frac{2\pi kS}{T}}{\pi k} = \frac{2}{T}\frac{\sin\omega S}{\omega}$$

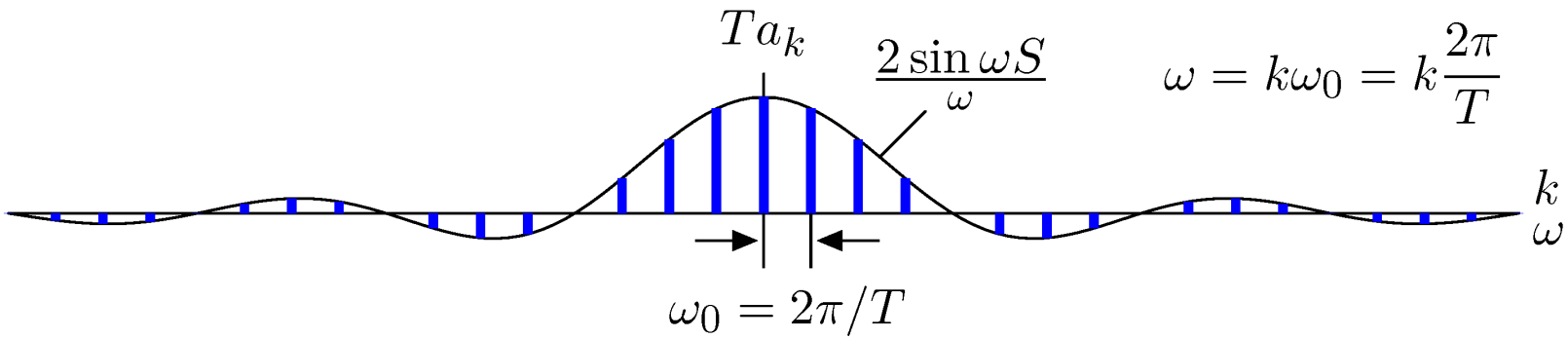


Fourier Transform

Doubling period doubles # of harmonics in given frequency interval.

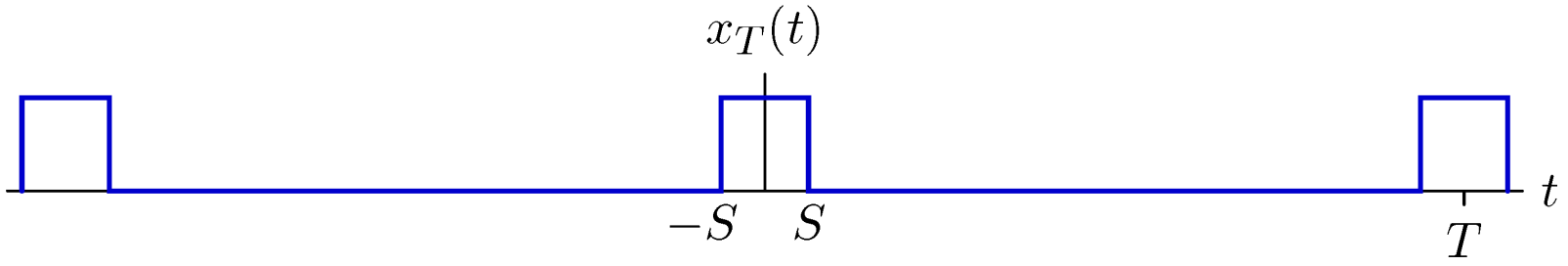


$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} x_T(t)e^{-j\frac{2\pi}{T}kt}dt = \frac{1}{T}\int_{-S}^{S} e^{-j\frac{2\pi}{T}kt}dt = \frac{\sin\frac{2\pi kS}{T}}{\pi k} = \frac{2}{T}\frac{\sin\omega S}{\omega}$$

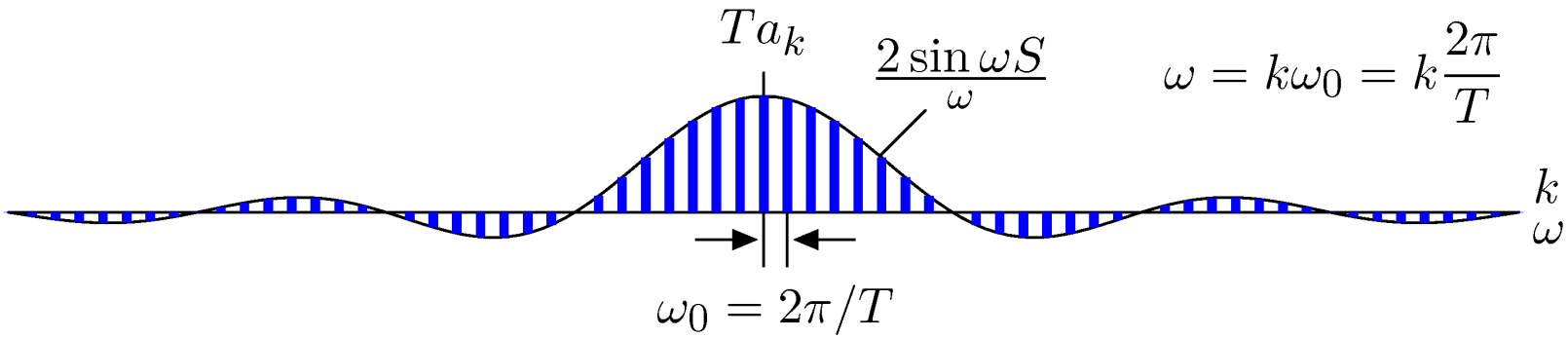


Fourier Transform

As $T \to \infty$, discrete harmonic amplitudes $\to$ a continuum $E(\omega)$.



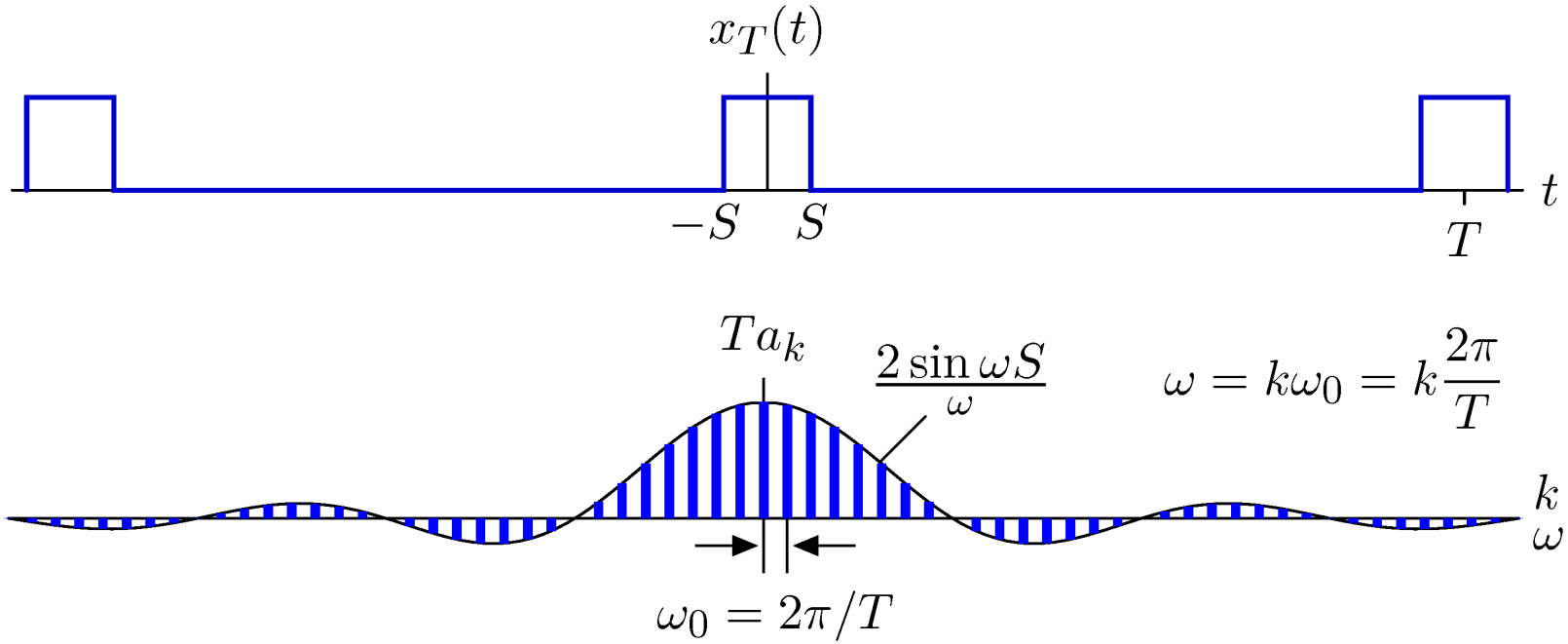
$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} x_T(t)e^{-j\frac{2\pi}{T}kt}dt = \frac{1}{T}\int_{-S}^{S} e^{-j\frac{2\pi}{T}kt}dt = \frac{\sin\frac{2\pi kS}{T}}{\pi k} = \frac{2}{T}\frac{\sin\omega S}{\omega}$$



$$\lim_{T\to\infty} Ta_k = \lim_{T\to\infty}\int_{-T/2}^{T/2} x(t)e^{-j\omega t}dt = \frac{2}{\omega}\sin\omega S = E(\omega)$$

Fourier Transform

As $T \to \infty$, synthesis sum $\to$ integral.



$$\lim_{T\to\infty} Ta_k = \lim_{T\to\infty} \int_{-T/2}^{T/2} x(t)e^{-j\omega t}dt = \frac{2}{\omega}\sin\omega S = E(\omega)$$

$$x(t) = \sum_{k=-\infty}^{\infty} \underbrace{\frac{1}{T}E(\omega)}_{a_k} e^{j\frac{2\pi}{T}kt} = \sum_{k=-\infty}^{\infty} \frac{\omega_0}{2\pi}E(\omega)e^{j\omega t} \to \frac{1}{2\pi}\int_{-\infty}^{\infty} E(\omega)e^{j\omega t}d\omega$$

Fourier Transform

Replacing $E(\omega)$ by $X(j\omega)$ yields the Fourier transform relations.

$$E(\omega) = X(s)|_{s=j\omega} \equiv X(j\omega)$$

Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \qquad \text{(“analysis” equation)}$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \qquad \text{(“synthesis” equation)}$$

Fourier Transform

Replacing $E(\omega)$ by $X(j\omega)$ yields the Fourier transform relations.

$$E(\omega) = X(s)|_{s=j\omega} \equiv X(j\omega)$$

Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \qquad \text{(“analysis” equation)}$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \qquad \text{(“synthesis” equation)}$$

Form is similar to that of Fourier series

$\rightarrow$ provides alternate view of signal.

Fourier Transform

Key Idea

Fourier series and transforms are similar: both represent signals by their frequency content.

Continuous-Time Fourier Transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \quad \text{(Synthesis equation)}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \quad \text{(Analysis equation)}$$

Continuous-Time Fourier Series

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi kt}{T}} \quad \text{(Synthesis equation)}$$

$$a_k = \frac{1}{T}\int_T x(t)e^{-j\frac{2\pi kt}{T}}dt \quad \text{(Analysis equation)}$$

where $\omega_0 = \frac{2\pi}{T}$

Fourier Transform

All of the information in a periodic signal is contained in one period.
The information in an aperiodic signal is spread across all time.

Continuous-Time Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t}\, d\omega \quad \text{Synthesis equation}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt \quad \text{Analysis equation}$$

Continuous-Time Fourier Series

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k \cdot e^{j\frac{2\pi kt}{T}} \quad \text{Synthesis equation}$$

$$a_k = \frac{1}{T} \int_{T} x(t) \cdot e^{-j\frac{2\pi kt}{T}}\, dt \quad \text{Analysis equation}$$

with $\omega_0 = \frac{2\pi}{T}$

Fourier Transform

Harmonic frequencies $k\omega_0$ are samples of continuous frequency ω

Continuous-Time Fourier Transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) \cdot e^{j\omega t}\, d\omega \quad \text{Synthesis equation}$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t}\, dt \quad \text{Analysis equation}$$

Continuous-Time Fourier Series

$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k \cdot e^{j\frac{2\pi kt}{T}} \quad \text{Synthesis equation}$$

$$a_k = \frac{1}{T} \int_{T} x(t) \cdot e^{-j\frac{2\pi kt}{T}}\, dt \quad \text{Analysis equation}$$

with $\omega_0 = \frac{2\pi}{T}$

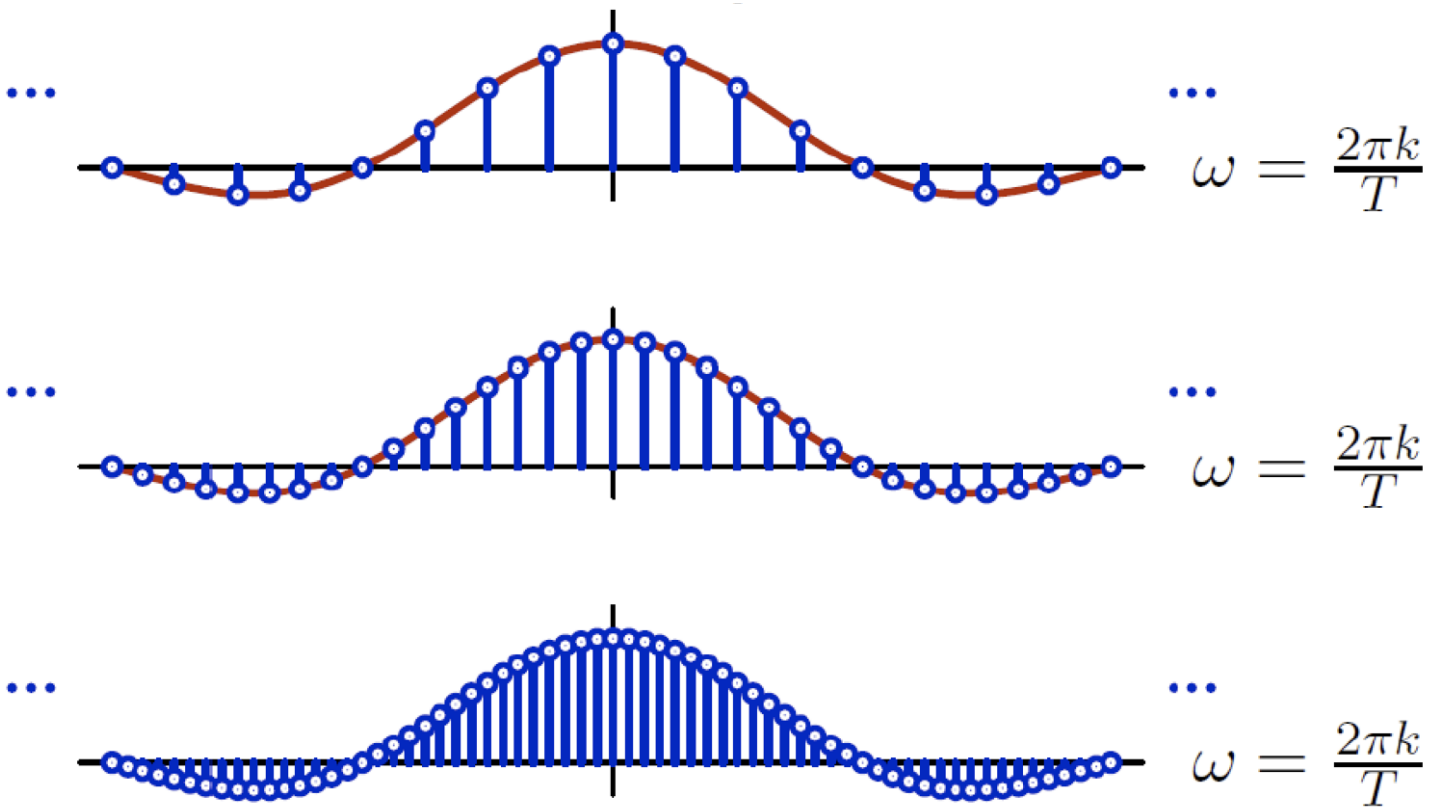
Fourier Transform

Harmonic frequencies $k\omega_0$ are samples of continuous frequency ω.

$$X(j\omega) = Ta_k = 2 \cdot \frac{\sin(\omega S)}{\omega}, \quad \text{where } \omega = k\omega_0 = \frac{2\pi k}{T}$$

Fourier Transform

- $T = 8$: Fewer samples, coarser resolution
- $T = 16$: More samples, better approximation
- $T = 32$: Dense sampling, closely follows continuous $X(j\omega)$



Check Yourself

Ex.#1 $x(t) = \delta(t)$

Check Yourself

Although physically unrealizable, the impulse (a.k.a. Dirac delta) function δ(t) is useful as a mathematically tractable approximation to a very brief signal.

δ(t) only has a nonzero value at t = 0, but it has finite area: it is most easily described as an integral:

$$\int_{-\infty}^{\infty} \delta(t)dt = \int_{0_-}^{0_+} \delta(t)dt = 1 \qquad \int_{-\infty}^{\infty} \delta(t-a)\ dt = \int_{a_-}^{a_+} \delta(t)\ dt = 1$$

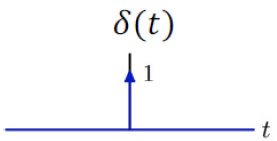
Importantly, it has the following property (the "sifting property"):

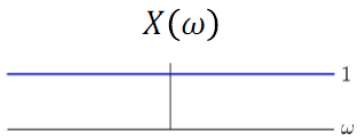
$$\int_{-\infty}^{\infty} \delta(t-a)f(t)dt = f(a)$$

$$let\ \tau = t-a,\ \int_{-\infty}^{\infty} \delta(\tau)f(\tau+a)d\tau = \int_{0_-}^{0_+} \delta(\tau)f(a)d\tau = f(a)\cdot\int_{0_-}^{0_+} \delta(\tau)d\tau = f(a)$$

The Fourier Transform of δ(t):

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t)\cdot e^{-j\omega t}\,dt = \int_{0_-}^{0_+} \delta(t)\cdot e^{-j\omega 0}\,dt \quad = 1$$





Check Yourself

Ex.#1 $x(t) = \delta(t)$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

$$\Downarrow$$

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega \qquad \text{— Synthesis equation for } \delta(t)$$

Check Yourself

Ex.#1 $x(t) = \delta(t)$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t) e^{-j\omega t} dt = 1$$

$$\Downarrow$$

$$\delta(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} d\omega \qquad \text{— Synthesis equation for } \delta(t)$$

Ex.#2 $x(t) = \delta(t - t_0)$

$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t - t_0) e^{-j\omega t} dt$$

$$= e^{-j\omega t_0}$$

Relation between Fourier and Laplace Transforms

If the Laplace transform of a signal exists and if the ROC includes the $j\omega$ axis, then the Fourier transform is equal to the Laplace transform evaluated on the $j\omega$ axis.

Laplace transform:

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$$

Fourier transform:

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = H(s)|_{s=j\omega}$$

Relation between Fourier and Laplace Transforms

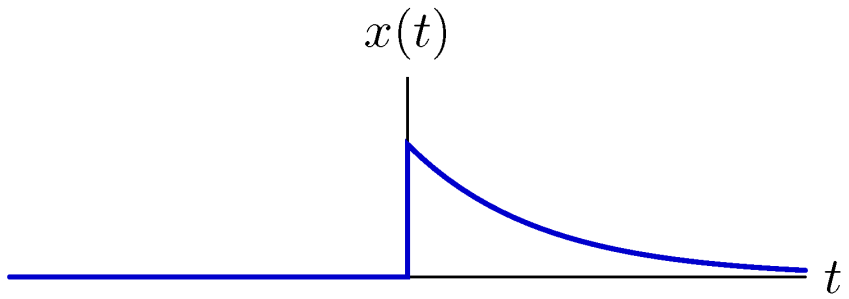
Fourier transform "inherits" properties of Laplace transform.

Property	$x(t)$	$X(s)$	$X(j\omega)$
Linearity	$ax_1(t) + bx_2(t)$	$aX_1(s) + bX_2(s)$	$aX_1(j\omega) + bX_2(j\omega$
Time shift	$x(t - t_0)$	$e^{-st_0}X(s)$	$e^{-j\omega t_0}X(j\omega)$
Time scale	$x(at)$	$\frac{1}{\lvert a\rvert}X\left(\frac{s}{a}\right)$	$\frac{1}{\lvert a\rvert}X\left(\frac{j\omega}{a}\right)$
Differentiation	$\frac{dx(t)}{dt}$	$sX(s)$	$j\omega X(j\omega)$
Multiply by t	$tx(t)$	$-\frac{d}{ds}X(s)$	$-\frac{1}{j}\frac{d}{d\omega}X(j\omega)$
Convolution	$x_1(t) * x_2(t)$	$X_1(s) \times X_2(s)$	$X_1(j\omega) \times X_2(j\omega)$

Relation between Fourier and Laplace Transforms

There are also important differences.

Compare Fourier and Laplace transforms of $x(t) = e^{-t}u(t)$.



Laplace transform

$$X(s) = \int_{-\infty}^{\infty} e^{-t}u(t)e^{-st}dt = \int_{0}^{\infty} e^{-(s+1)t}dt = \frac{1}{1+s} \;;\; \mathrm{Re}(s) > -1$$

a complex-valued function of **complex** domain.

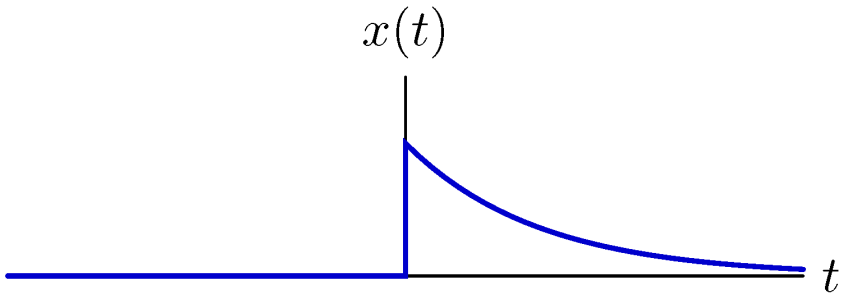
Fourier transform

$$X(j\omega) = \int_{-\infty}^{\infty} e^{-t}u(t)e^{-j\omega t}dt = \int_{0}^{\infty} e^{-(j\omega+1)t}dt = \frac{1}{1+j\omega}$$

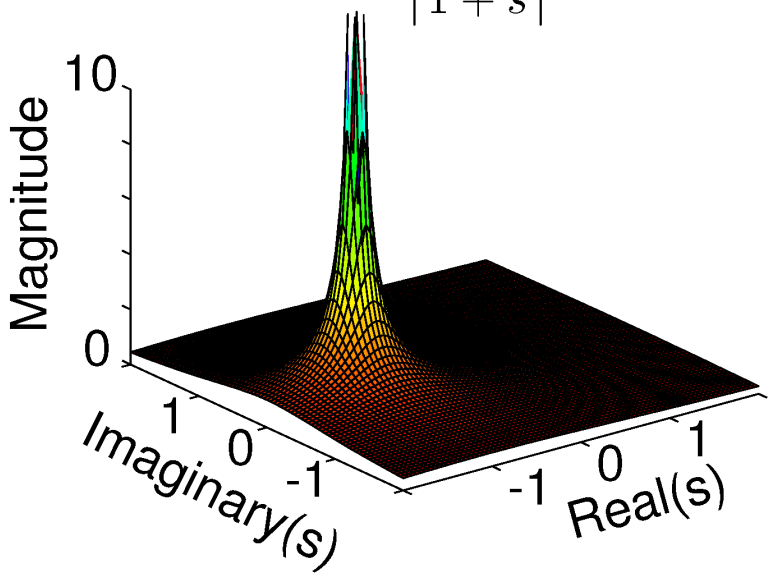
a complex-valued function of **real** domain.

Laplace Transform

The Laplace transform maps a function of time t to a complex-valued function of complex-valued domain s.

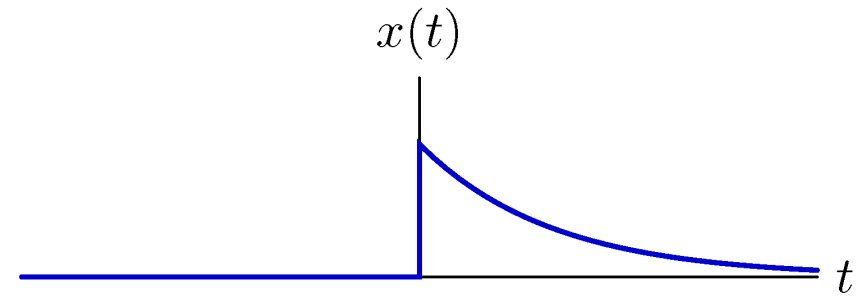


$$|X(s)| = \left| \frac{1}{1+s} \right|$$

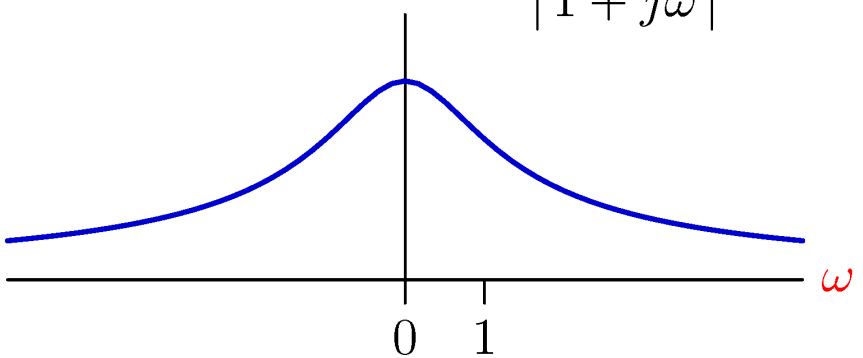


Fourier Transform

The Fourier transform maps a function of time t to a complex-valued function of real-valued domain ω.

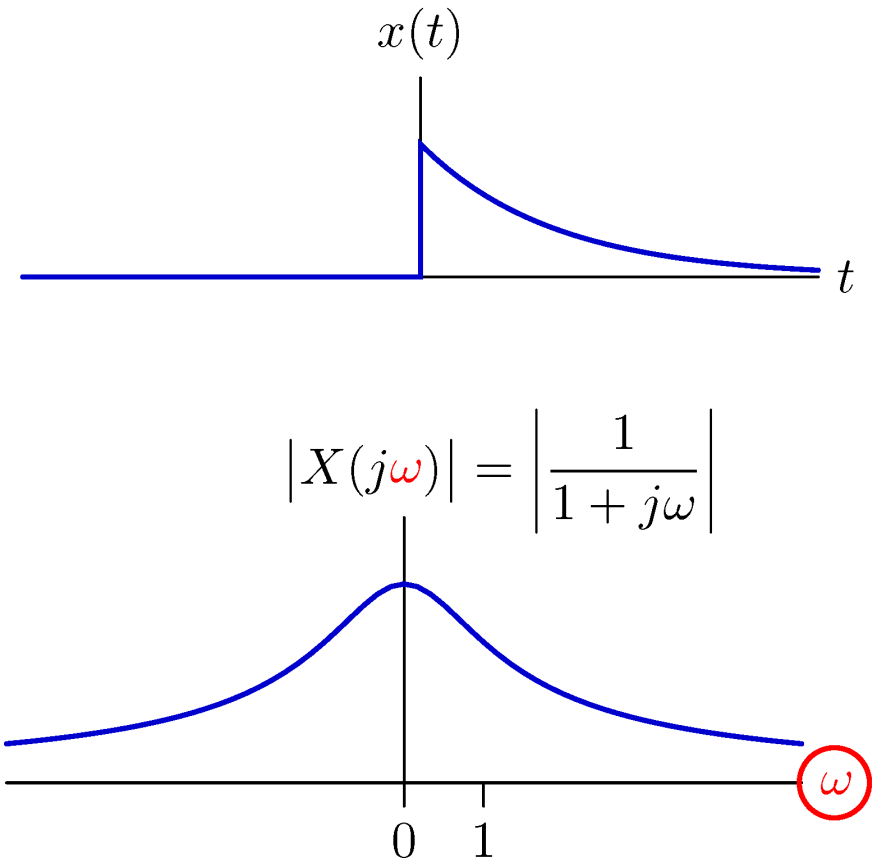


$$|X(j\omega)| = \left|\frac{1}{1+j\omega}\right|$$



Fourier Transform

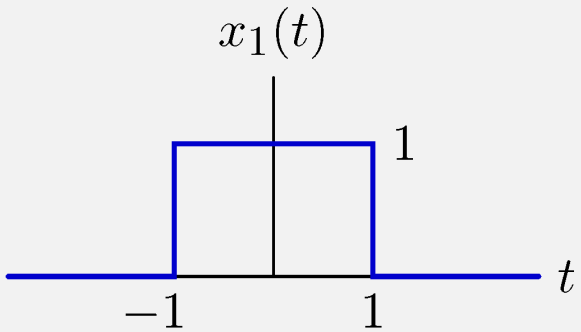
The Fourier transform maps a function of time t to a complex-valued function of real-valued domain ω.



Frequency plots provide intuition that is difficult to otherwise obtain.

Check Yourself

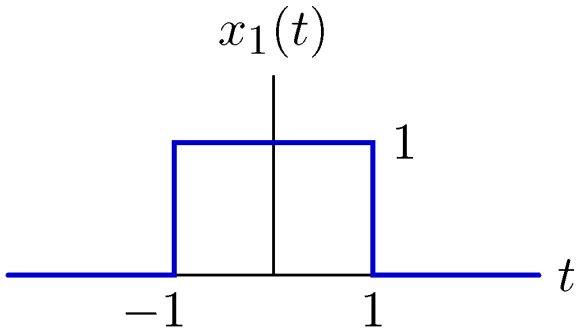
Find the Fourier transform of the following square pulse.



1. $X_1(j\omega) = \dfrac{1}{\omega}\left(e^{\omega} - e^{-\omega}\right)$
2. $X_1(j\omega) = \dfrac{1}{\omega}\sin\omega$
3. $X_1(j\omega) = \dfrac{2}{\omega}\left(e^{\omega} - e^{-\omega}\right)$
4. $X_1(j\omega) = \dfrac{2}{\omega}\sin\omega$
5. none of the above

Fourier Transform

Compare the Laplace and Fourier transforms of a square pulse.



Laplace transform:

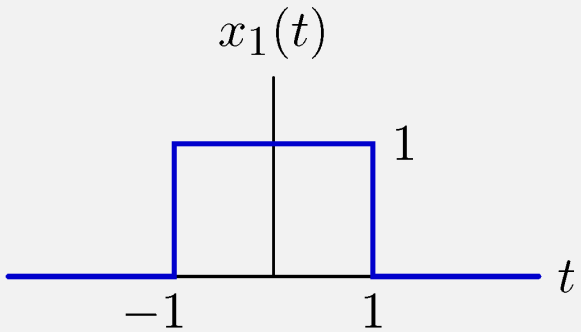
$$X_1(s) = \int_{-1}^{1} e^{-st} dt = \frac{e^{-st}}{-s}\Big|_{-1}^{1} = \frac{1}{s}\left(e^{s} - e^{-s}\right) \quad \text{[function of } s = \sigma + j\omega\text{]}$$

Fourier transform

$$X_1(j\omega) = \int_{-1}^{1} e^{-j\omega t} dt = \frac{e^{-j\omega t}}{-j\omega}\Big|_{-1}^{1} = \frac{2\sin\omega}{\omega} \quad \text{[function of } \omega\text{]}$$

Check Yourself

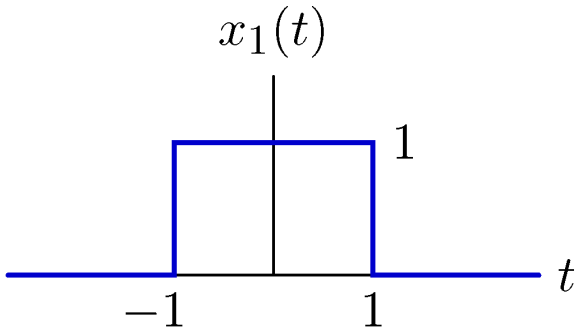
Find the Fourier transform of the following square pulse. 4



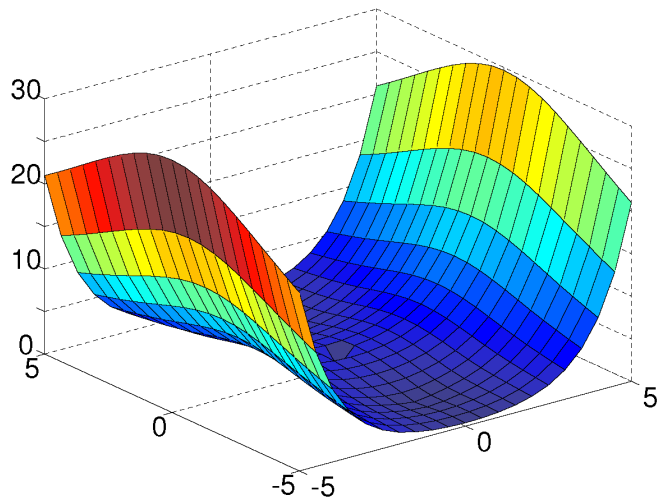
1. $X_1(j\omega) = \dfrac{1}{\omega}\left(e^{\omega} - e^{-\omega}\right)$
2. $X_1(j\omega) = \dfrac{1}{\omega}\sin\omega$
3. $X_1(j\omega) = \dfrac{2}{\omega}\left(e^{\omega} - e^{-\omega}\right)$
4. $X_1(j\omega) = \dfrac{2}{\omega}\sin\omega$
5. none of the above

Laplace Transform

Laplace transform: complex-valued function of complex domain.



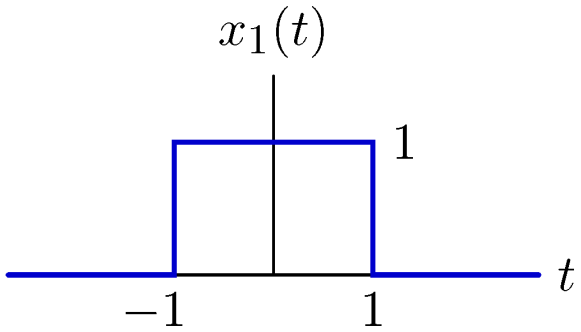
$$|X(s)| = \left|\frac{1}{s}(e^s - e^{-s})\right|$$



Fourier Transform

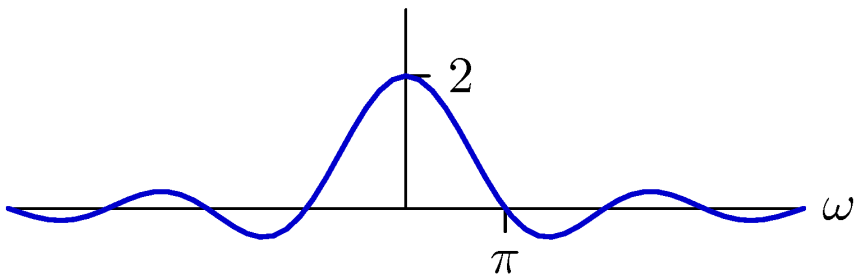
The Fourier transform is a function of real domain: frequency ω.

Time representation:



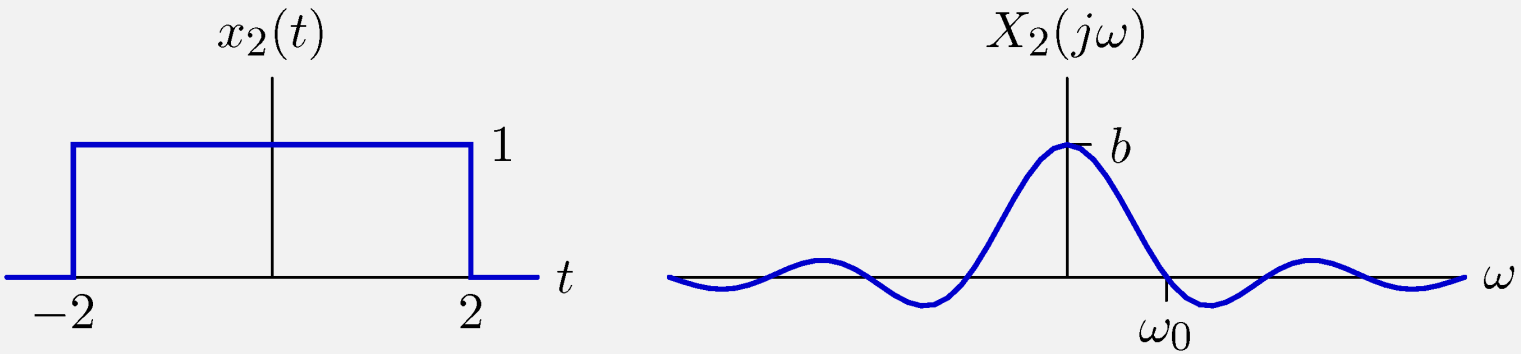
Frequency representation:

$$X_1(j\omega) = \frac{2\sin\omega}{\omega}$$



Check Yourself

Signal $x_2(t)$ and its Fourier transform $X_2(j\omega)$ are shown below.



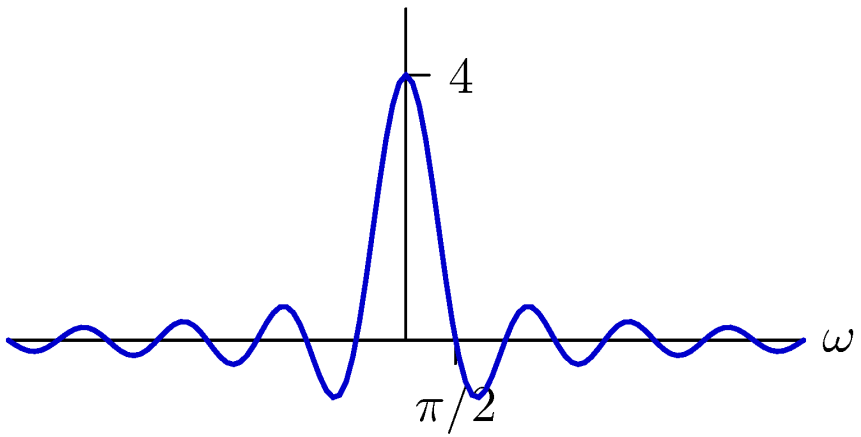
Which is true?

1. $b = 2$ and $\omega_0 = \pi/2$
2. $b = 2$ and $\omega_0 = 2\pi$
3. $b = 4$ and $\omega_0 = \pi/2$
4. $b = 4$ and $\omega_0 = 2\pi$
5. none of the above

Check Yourself

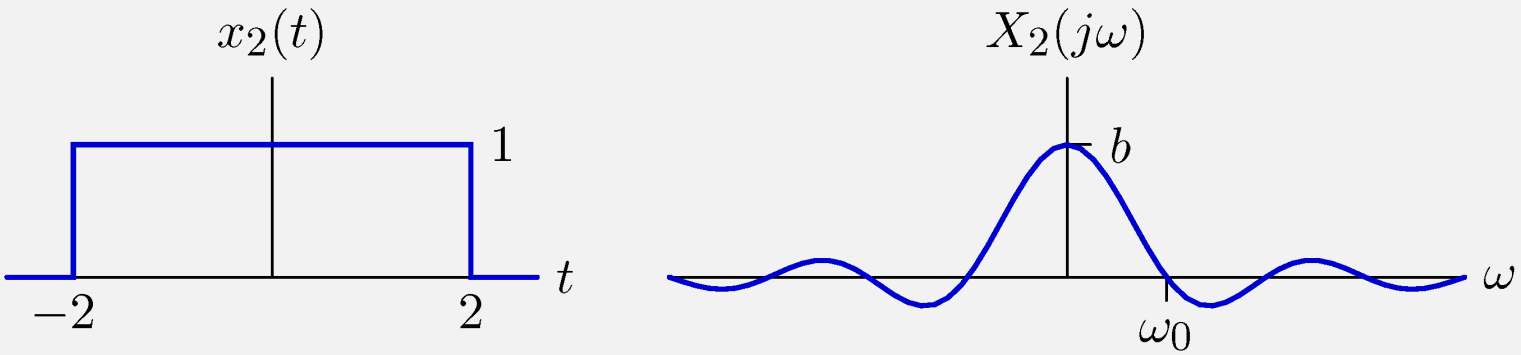
Find the Fourier transform.

$$X_2(j\omega) = \int_{-2}^{2} e^{-j\omega t} dt = \frac{e^{-j\omega t}}{-j\omega}\bigg|_{-2}^{2} = \frac{2\sin 2\omega}{\omega} = \frac{4\sin 2\omega}{2\omega}$$



Check Yourself

Signal $x_2(t)$ and its Fourier transform $X_2(j\omega)$ are shown below.

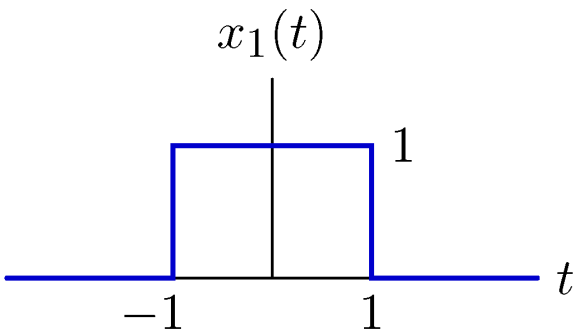


Which is true? 3

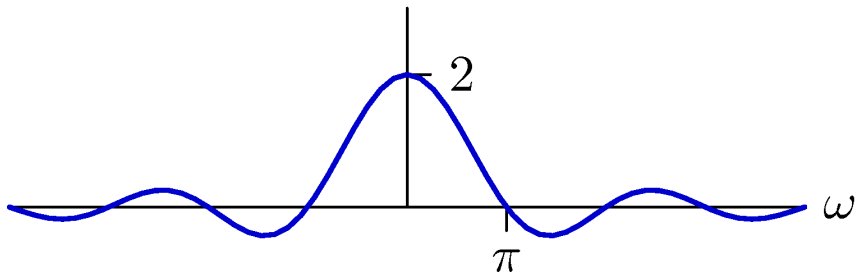
1. $b = 2$ and $\omega_0 = \pi/2$
2. $b = 2$ and $\omega_0 = 2\pi$
3. $b = 4$ and $\omega_0 = \pi/2$
4. $b = 4$ and $\omega_0 = 2\pi$
5. none of the above

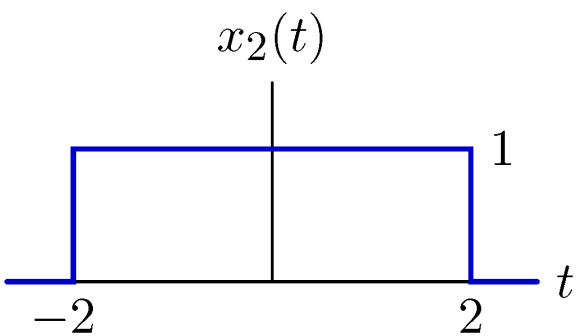
Fourier Transforms

Stretching time compresses frequency.

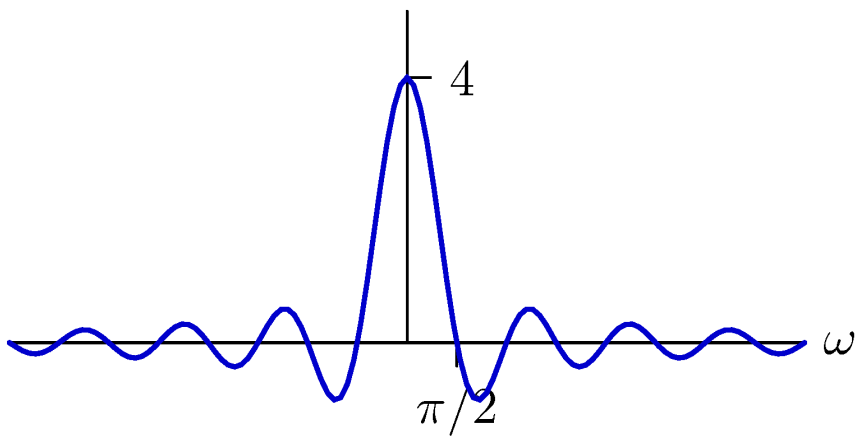


$$X_1(j\omega) = \frac{2\sin\omega}{\omega}$$





$$X_2(j\omega) = \frac{4\sin 2\omega}{2\omega}$$



Check Yourself

Stretching time compresses frequency.

Find a general scaling rule.

Let $x_2(t) = x_1(at)$.

If time is stretched in going from x_1 to x_2, is $a > 1$ or $a < 1$?

Check Yourself

Stretching time compresses frequency.

Find a general scaling rule.

Let $x_2(t) = x_1(at)$.

If time is stretched in going from x_1 to x_2, is $a > 1$ or $a < 1$?

$$x_2(2) = x_1(1)$$

$$x_2(t) = x_1(at)$$

Therefore $a = 1/2$, or more generally, $a < 1$.

Check Yourself

Stretching time compresses frequency.

Find a general scaling rule.

Let $x_2(t) = x_1(at)$.

If time is stretched in going from x_1 to x_2, is $a > 1$ or $a < 1$?

$a < 1$

Fourier Transforms

Find a general scaling rule.

Let $x_2(t) = x_1(at)$.

$$X_2(j\omega) = \int_{-\infty}^{\infty} x_2(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} x_1(at)e^{-j\omega t}dt$$

Let $\tau = at$ $(a > 0)$.

$$X_2(j\omega) = \int_{-\infty}^{\infty} x_1(\tau)e^{-j\omega\tau/a}\ \frac{1}{a}\,d\tau = \frac{1}{a}X_1\left(\frac{j\omega}{a}\right)$$

If $a < 0$ the sign of $d\tau$ would change along with the limits of integration. In general,

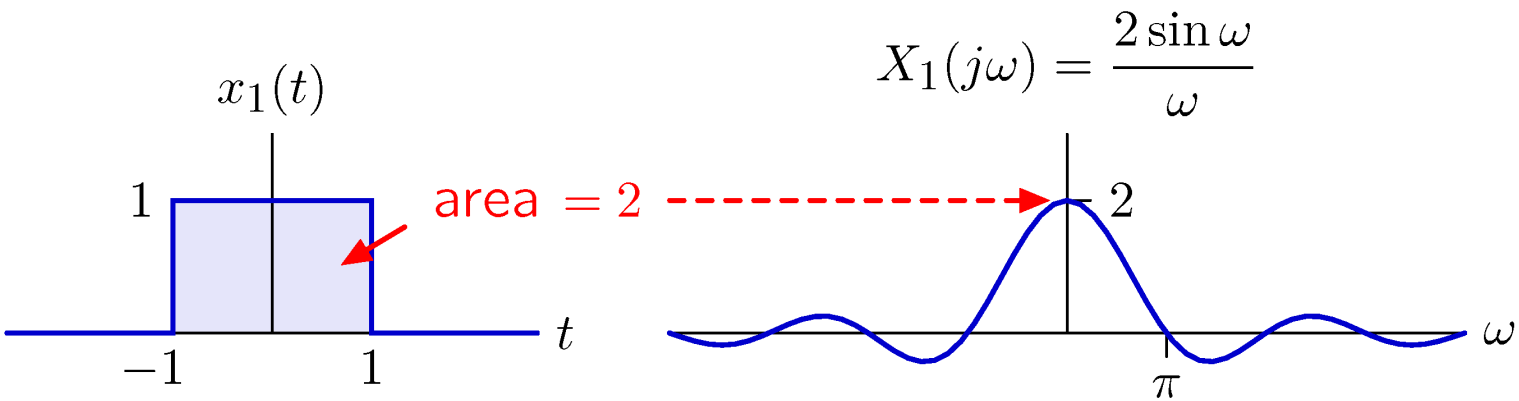
$$x_1(at) \ \leftrightarrow\ \frac{1}{|a|}X_1\left(\frac{j\omega}{a}\right).$$

If time is stretched $(a < 1)$ then frequency is compressed and amplitude increases (preserving area).

Moments

The value of $X(j\omega)$ at $\omega = 0$ is the integral of $x(t)$ over time t.

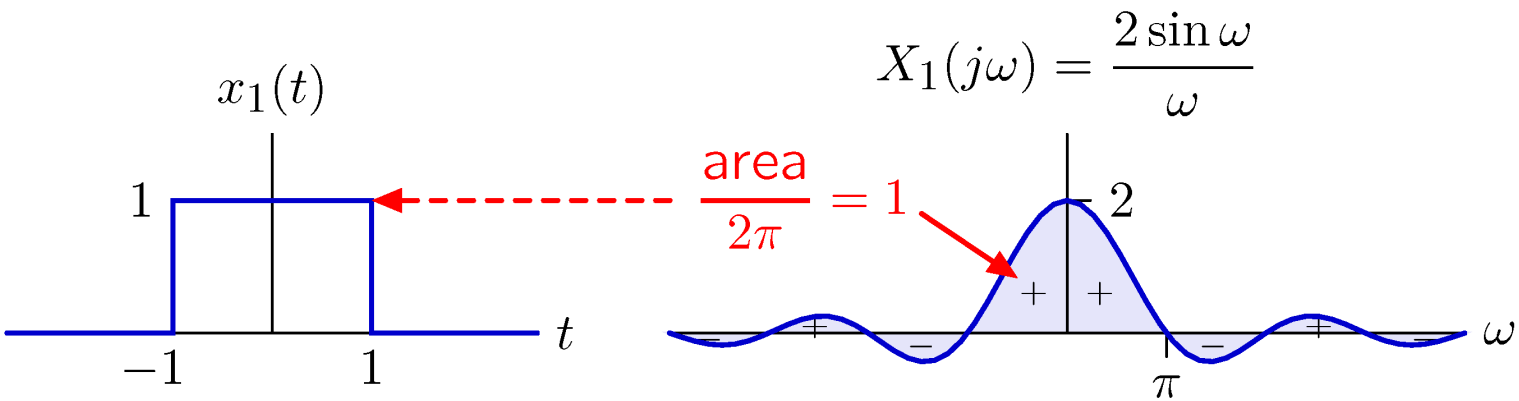
$$X(j\omega)|_{\omega=0} = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt = \int_{-\infty}^{\infty} x(t)e^{j0t}dt = \int_{-\infty}^{\infty} x(t)\,dt$$



Moments

The value of $x(0)$ is the integral of $X(j\omega)$ divided by 2π.

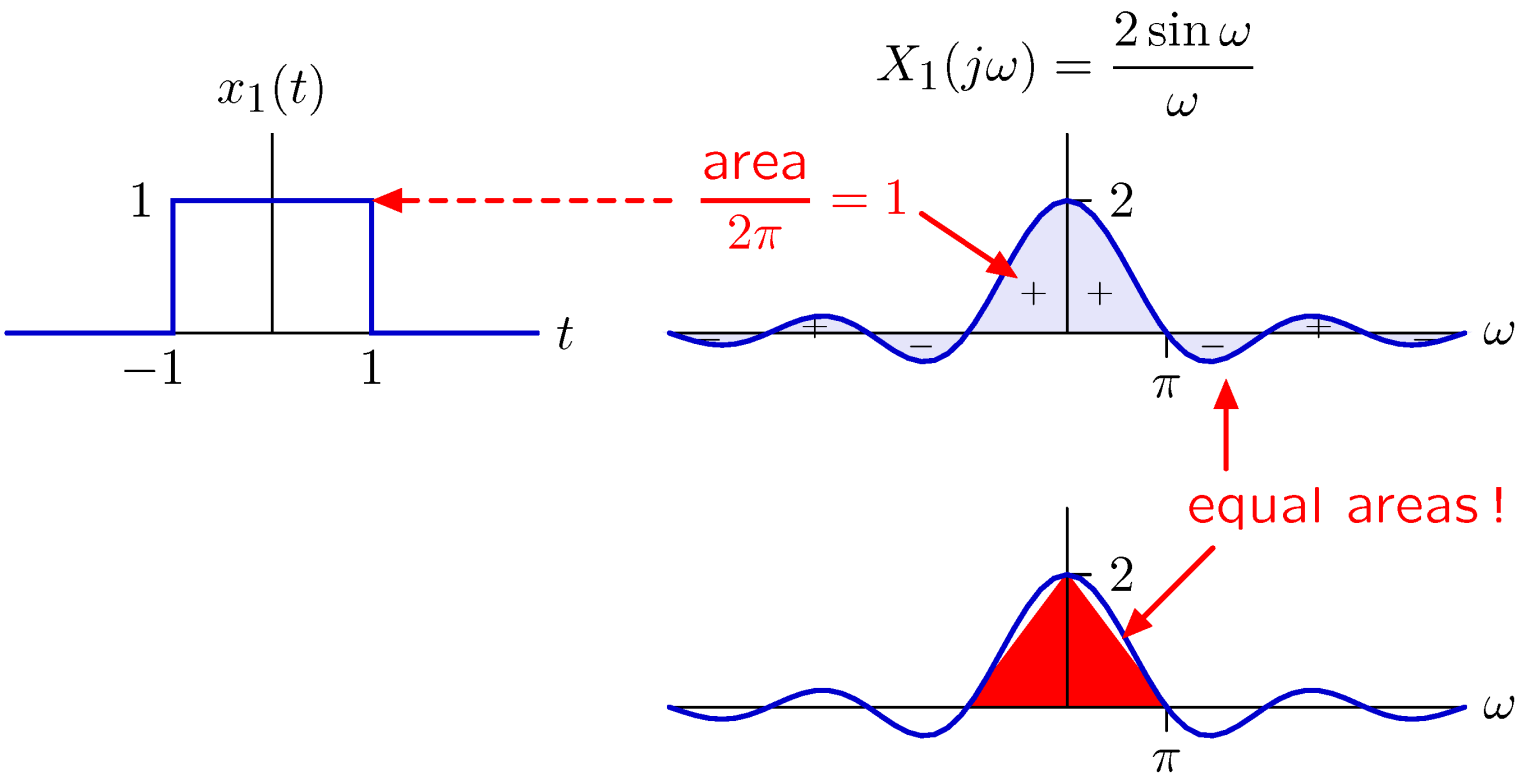
$$x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t} d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, d\omega$$



Moments

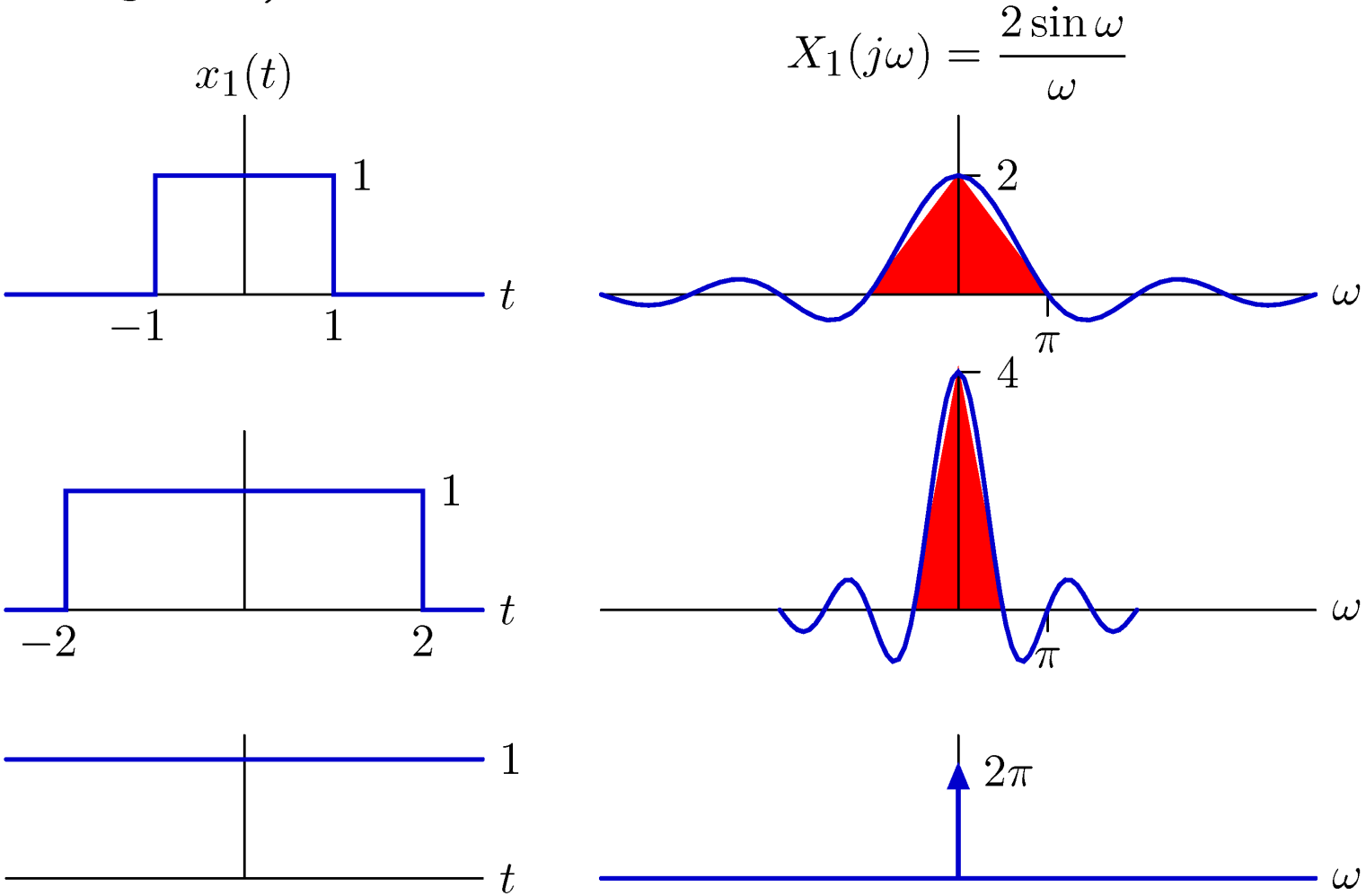
The value of $x(0)$ is the integral of $X(j\omega)$ divided by 2π.

$$x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t} d\omega = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, d\omega$$



Stretching to the Limit

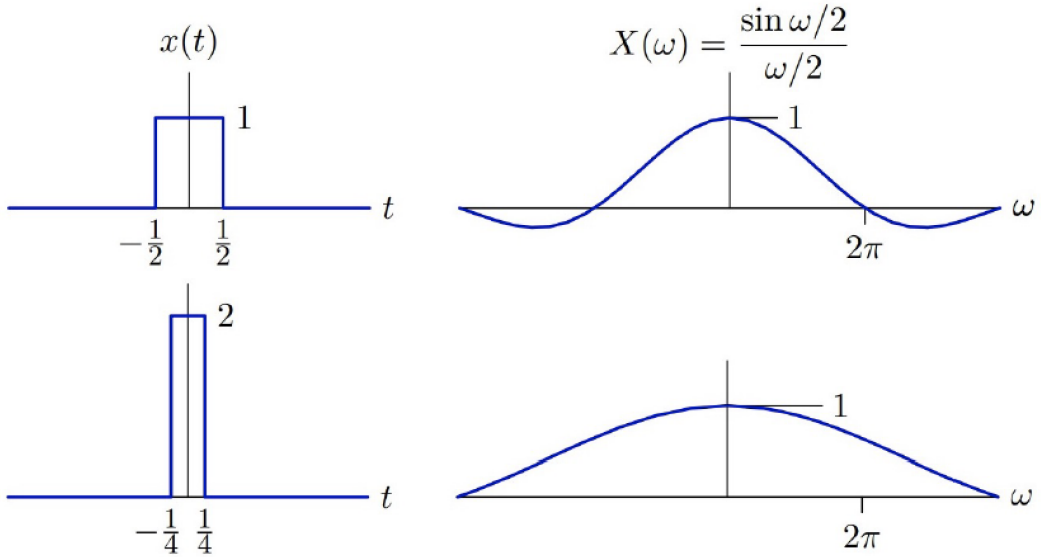
Stretching time compresses frequency and increases amplitude (preserving area).



New way to think about an impulse!

Compressing Time to the Limit

Alternatively, compress time while keeping area = 1:



In the limit, the pulse has zero width but area 1! We represent this limit with the delta function: δ(t)



Fourier Transform

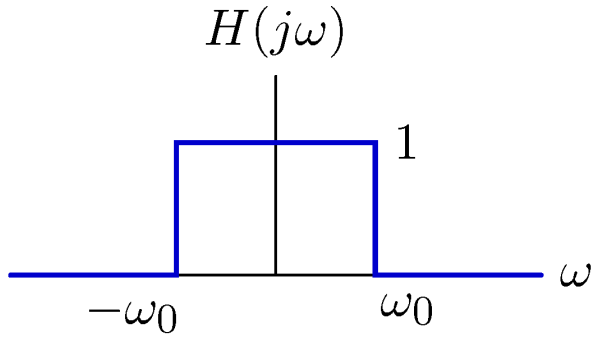
One of the most useful features of the Fourier transform (and Fourier series) is the simple "inverse" Fourier transform.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \qquad \text{(Fourier transform)}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \qquad \text{("inverse" Fourier transform)}$$

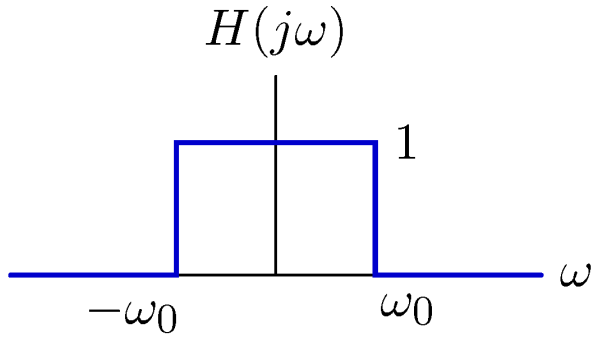
Inverse Fourier Transform

Find the impulse reponse of an "ideal" low pass filter.

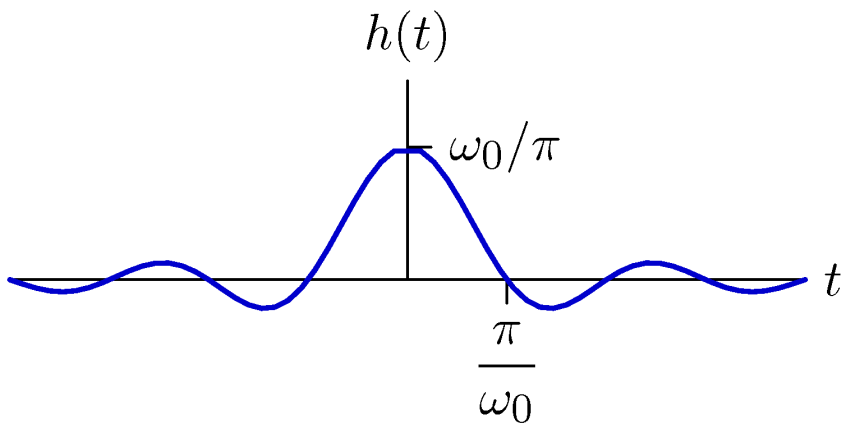


Inverse Fourier Transform

Find the impulse reponse of an "ideal" low pass filter.



$$h(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} H(j\omega)e^{j\omega t}d\omega = \frac{1}{2\pi}\int_{-\omega_0}^{\omega_0} e^{j\omega t}d\omega = \frac{1}{2\pi}\left.\frac{e^{j\omega t}}{jt}\right|_{-\omega_0}^{\omega_0} = \frac{\sin\omega_0 t}{\pi t}$$



This result is not so easily obtained without inverse relation.

Fourier Transform of a Unit Impulse

Find the function whose Fourier transform is a unit impulse.

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \delta(\omega)\cdot e^{j\omega t}\,d\omega \;= \frac{1}{2\pi}\int_{0_-}^{0_+} \delta(\omega)\cdot e^{j0t}\,d\omega \;= \frac{1}{2\pi}$$

$$1 \quad \overset{\text{CTFT}}{\Longleftrightarrow} \quad 2\pi\delta(\omega)$$

Notice the similarity to the previous result:

$$\delta(t) \quad \overset{\text{CTFT}}{\Longleftrightarrow} \quad 1$$

These relations are **duals** of each other:

- A constant in time consists of a single frequency at ω = 0.
- An impulse in time contains components at all frequencies.

Fourier Transform

The Fourier transform and its inverse have very similar forms.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \qquad \text{(Fourier transform)}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \qquad \text{(“inverse” Fourier transform)}$$

Convert one to the other by

- $t \to \omega$
- $\omega \to -t$
- scale by 2π

Duality

The Fourier transform and its inverse have very similar forms.

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$$

Two differences:

- minus sign: flips time axis (or equivalently, frequency axis)
- divide by 2π (or multiply in the other direction)

$$x_1(t) = f(t) \leftrightarrow X_1(j\omega) = g(\omega)$$

$\omega \to t$ $\quad t \to \omega$; flip ; $\times 2\pi$

$$x_2(t) = g(t) \leftrightarrow X_2(j\omega) = 2\pi f(-\omega)$$

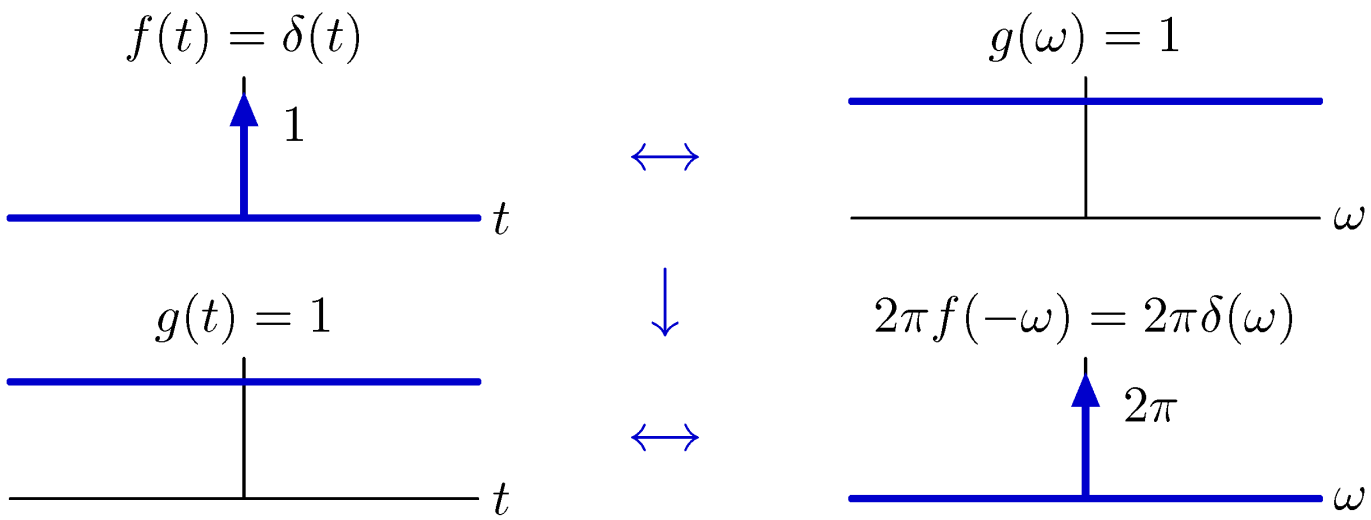
Duality

Using duality to find new transform pairs.

$$x_1(t) = f(t) \leftrightarrow X_1(j\omega) = g(\omega)$$

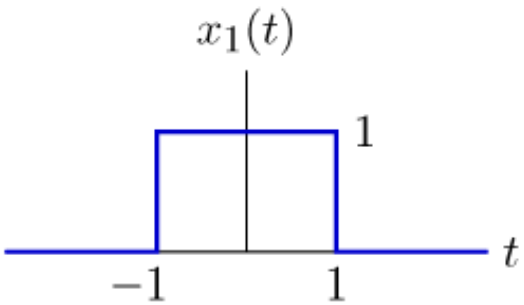
$\omega \to t$ $\quad t \to \omega$; flip ; $\times 2\pi$

$$x_2(t) = g(t) \leftrightarrow X_2(j\omega) = 2\pi f(-\omega)$$

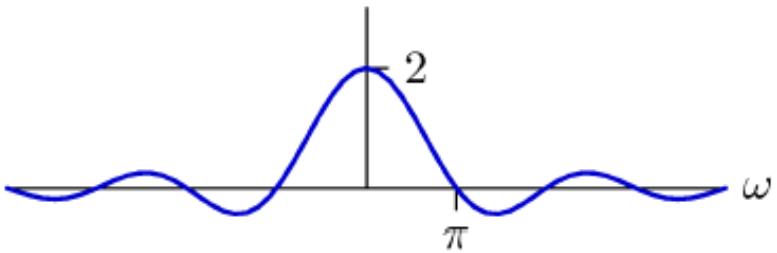


The function $g(t) = 1$ does not have a Laplace transform!

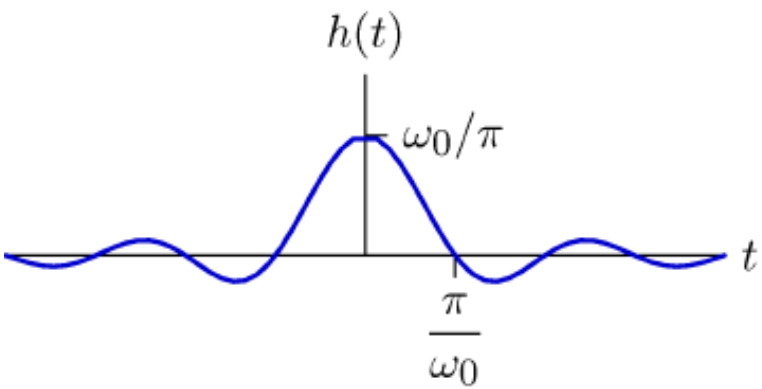
Duality

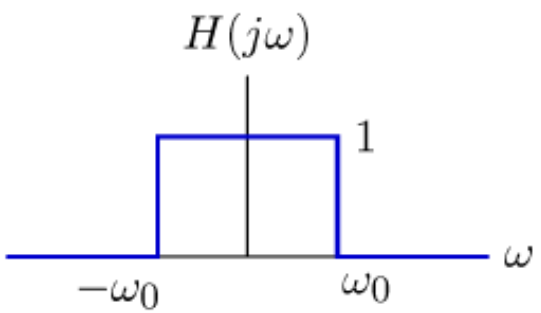


$$X_1(j\omega) = \frac{2\sin\omega}{\omega}$$

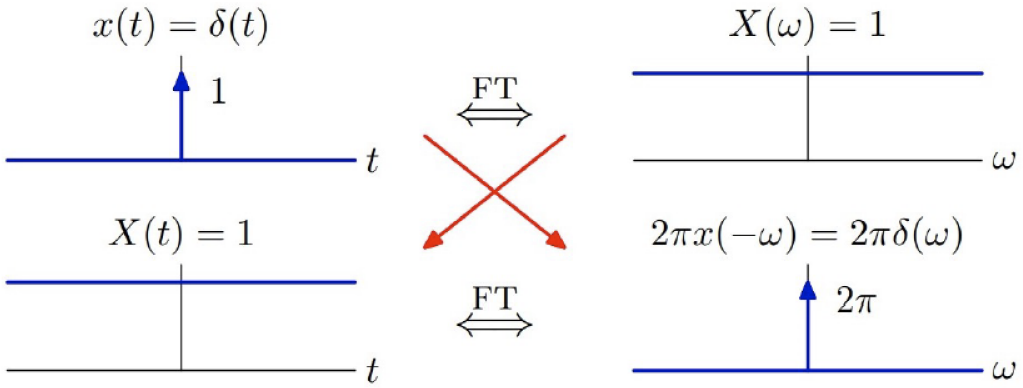




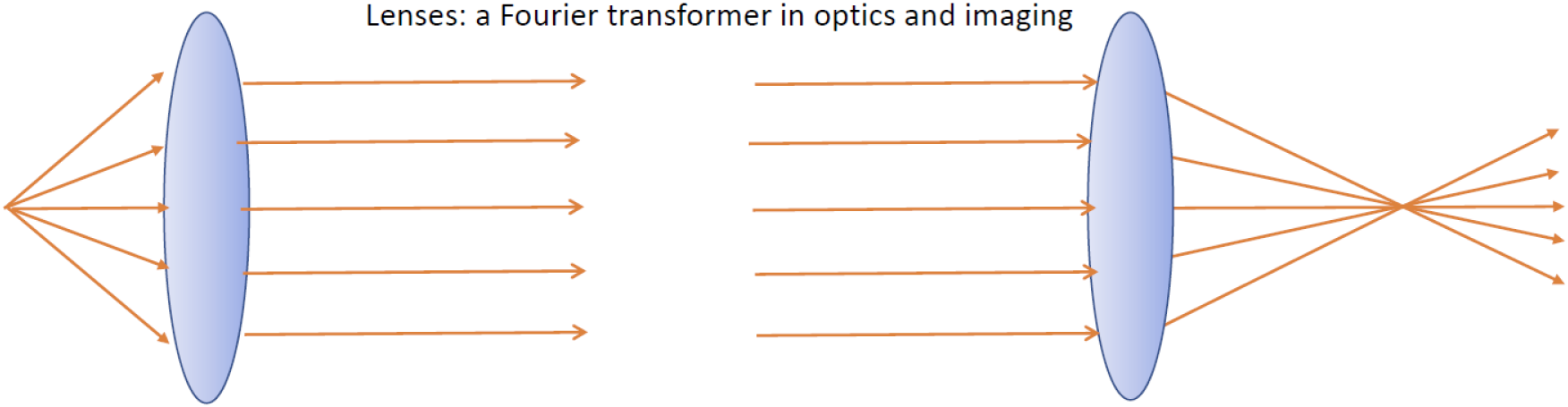




Duality in Optics/Imaging/Photography



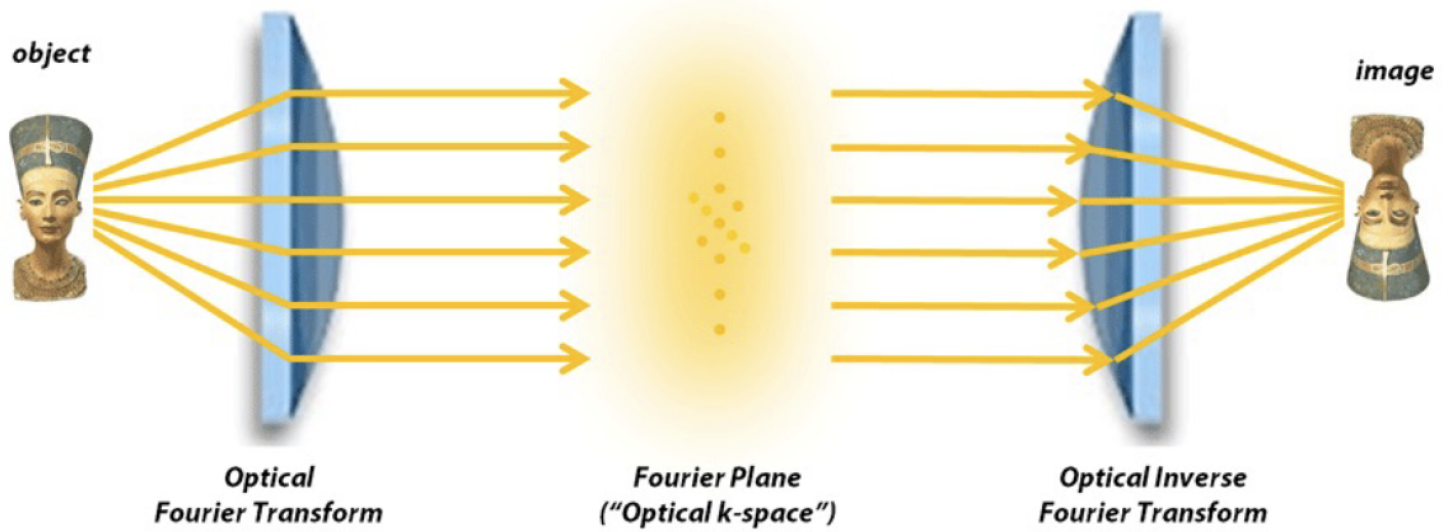
Lenses: a Fourier transformer in optics and imaging



The Principle of Reversibility: the direction of the light paths is always reversible

Duality in Optics/Imaging/Photography

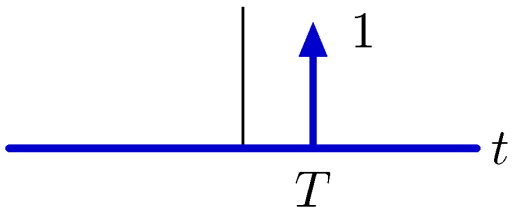
The FT of a FT of a signal = a scaled flipped signal



More Impulses

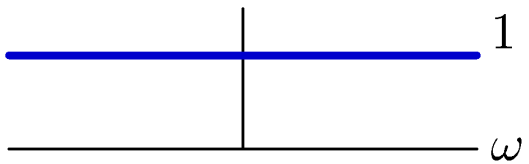
Fourier transform of delayed impulse: $\delta(t-T) \leftrightarrow e^{-j\omega T}$.

$$x(t) = \delta(t-T)$$

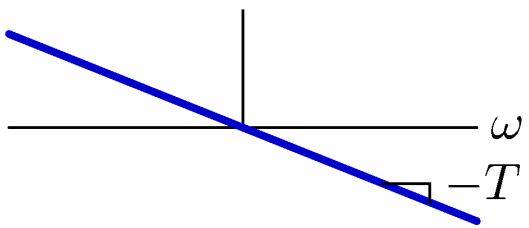


$$X(j\omega) = \int_{-\infty}^{\infty} \delta(t-T)e^{-j\omega t}dt = e^{-j\omega T}$$

$$\left|X(j\omega)\right| = 1$$



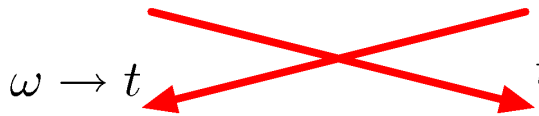
$$\angle X(j\omega) = -\omega T$$



Eternal Sinusoids

Using duality to find the Fourier transform of an eternal sinusoid.

$$\delta(t-T) \quad \leftrightarrow \quad e^{-j\omega T}$$



$\omega \to t$ $\qquad$ $t \to \omega$; flip ; $\times 2\pi$

$$e^{-jtT} \quad \leftrightarrow \quad 2\pi\delta(\omega+T)$$

$T \to \omega_0$:

$$e^{-j\omega_0 t} \quad \leftrightarrow \quad 2\pi\delta(\omega+\omega_0)$$

Check Yourself

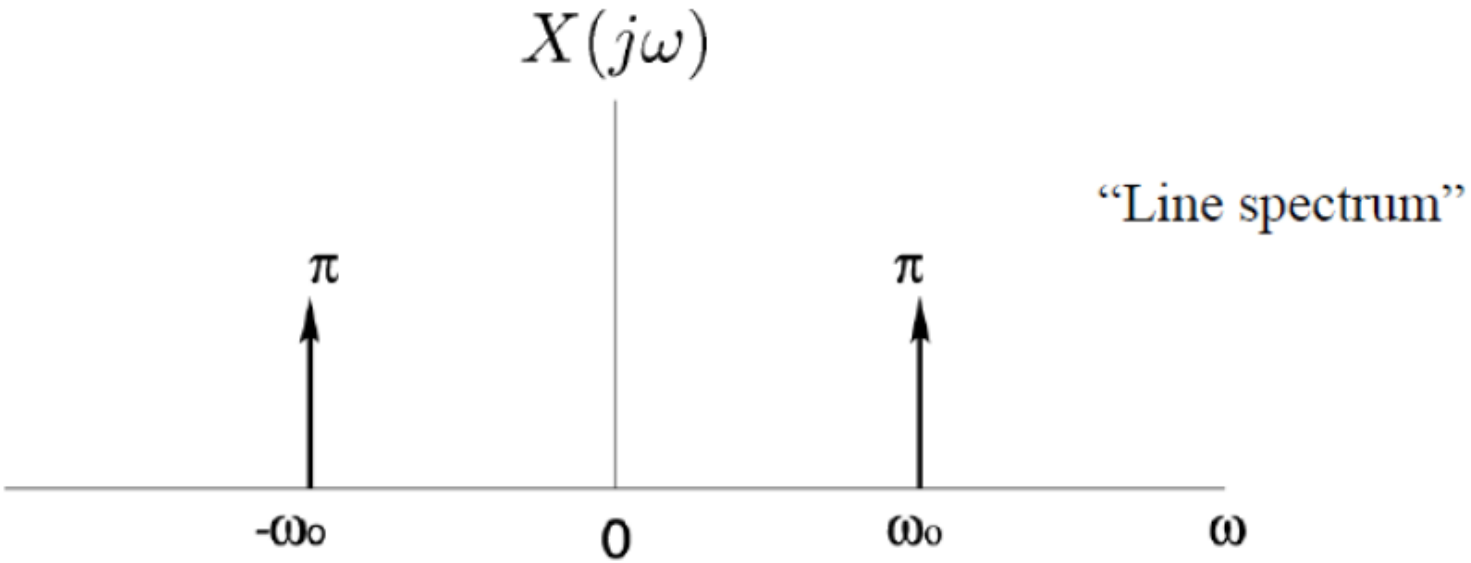
$$x(t) = \cos \omega_0 t = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$$

Check Yourself

$$x(t) = \cos \omega_0 t = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$$

$$\updownarrow$$

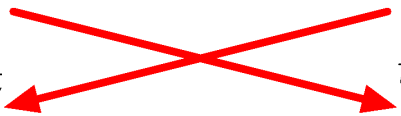
$$X(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$



Eternal Sinusoids

Using duality to find the Fourier transform of an eternal sinusoid.

$$\delta(t-T) \quad \leftrightarrow \quad e^{-j\omega T}$$



$\omega \to t$ $\qquad$ $t \to \omega$; flip ; $\times 2\pi$

$$e^{-jtT} \quad \leftrightarrow \quad 2\pi\delta(\omega+T)$$

$T \to \omega_0$:

$$e^{-j\omega_0 t} \quad \leftrightarrow \quad 2\pi\delta(\omega+\omega_0)$$

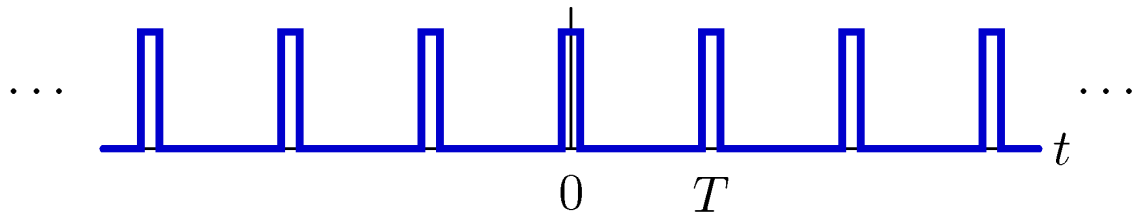
$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt} \quad \overset{\text{CTFS}}{\longleftrightarrow} \quad \{a_k\}$$

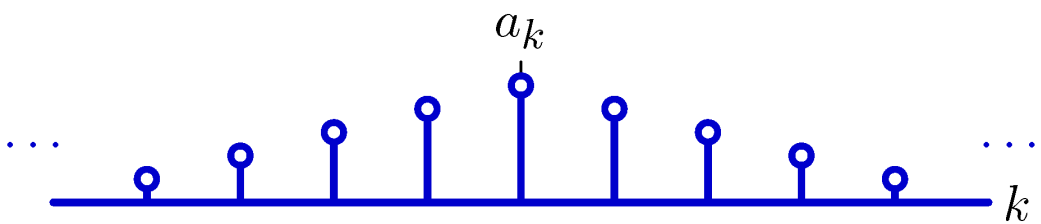
$$x(t) = x(t+T) = \sum_{k=-\infty}^{\infty} a_k e^{j\frac{2\pi}{T}kt} \quad \overset{\text{CTFT}}{\longleftrightarrow} \quad \sum_{k=-\infty}^{\infty} 2\pi a_k \delta\left(\omega - \frac{2\pi}{T}k\right)$$

Relation between Fourier Transform and Fourier Series

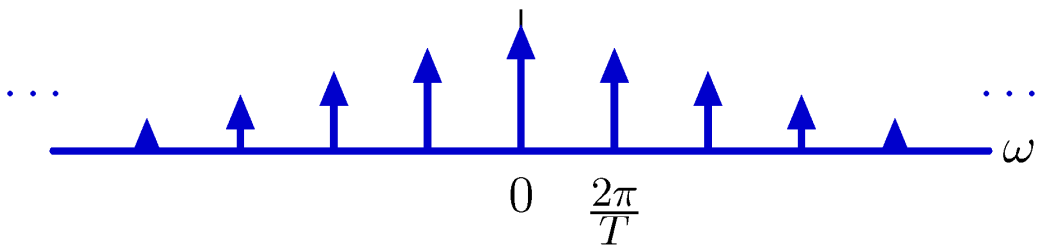
Each term in the Fourier series is replaced by an impulse.

$$x(t) = \sum_{k=-\infty}^{\infty} x_p(t - kT)$$





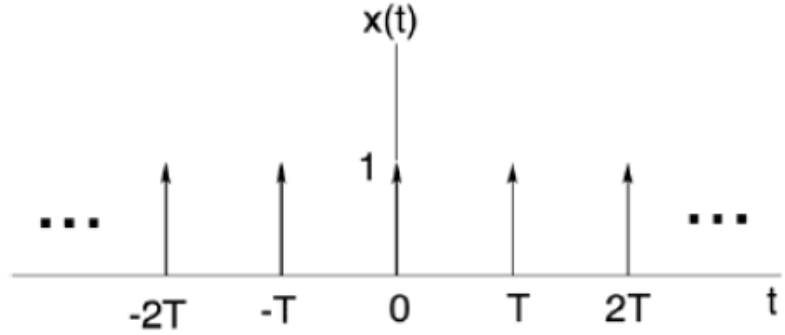
$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi\, a_k\, \delta(\omega - k\frac{2\pi}{T})$$



Check Yourself

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

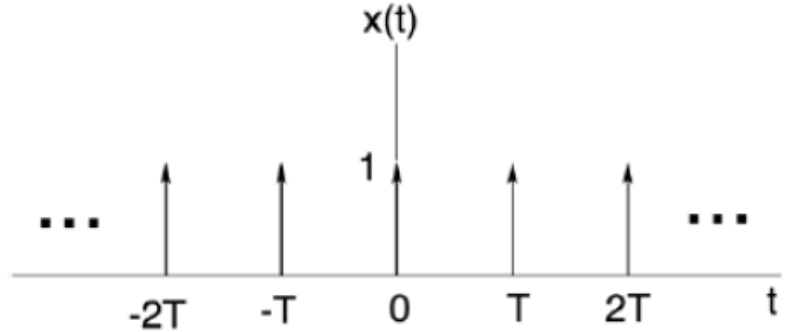
— Sampling function



Check Yourself

$$x(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$$

— Sampling function

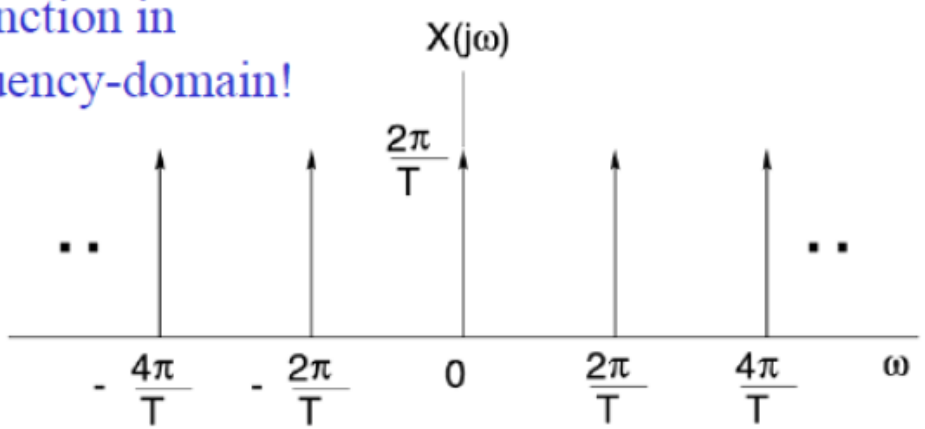


$$x(t) \leftrightarrow a_k = \frac{1}{T}\int_{-T/2}^{T/2} x(t)e^{-jk\omega_0 t}dt = \frac{1}{T}$$

$$\Downarrow$$

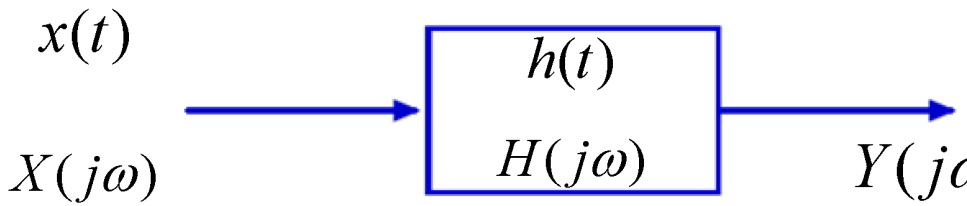
$$X(j\omega) = \sum_{n=-\infty}^{\infty} \underbrace{\frac{2\pi}{T}}_{2\pi a_k} \delta(\omega - \underbrace{\frac{k2\pi}{T}}_{k\omega_0})$$

Same function in the frequency-domain!



Note: (period in t) T
$\Leftrightarrow$ (period in ω) $2\pi/T$
Inverse relationship again!

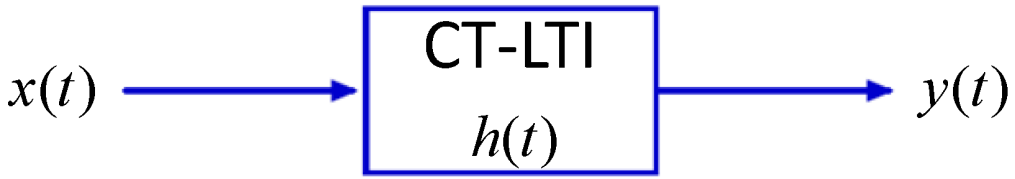
The convolution property



$$
\begin{aligned}
Y(j\omega) &= \int_{-\infty}^{\infty} y(t) e^{-j\omega t} dt \\
&= \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau \right] e^{-j\omega t} dt \\
&= \int_{-\infty}^{\infty} x(\tau) \left[\int_{-\infty}^{\infty} h(t-\tau) e^{-j\omega t} dt \right] d\tau \\
&= \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} H(j\omega) d\tau \\
&= H(j\omega) \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} d\tau = H(j\omega) X(j\omega)
\end{aligned}
$$

CT Fourier Transform

The Frequency Response



$$x(t) = e^{j\omega t} \longrightarrow y(t) \ ?$$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)d\tau$$

$$= \int_{-\infty}^{\infty} e^{j\omega(t-\tau)}h(\tau)d\tau$$

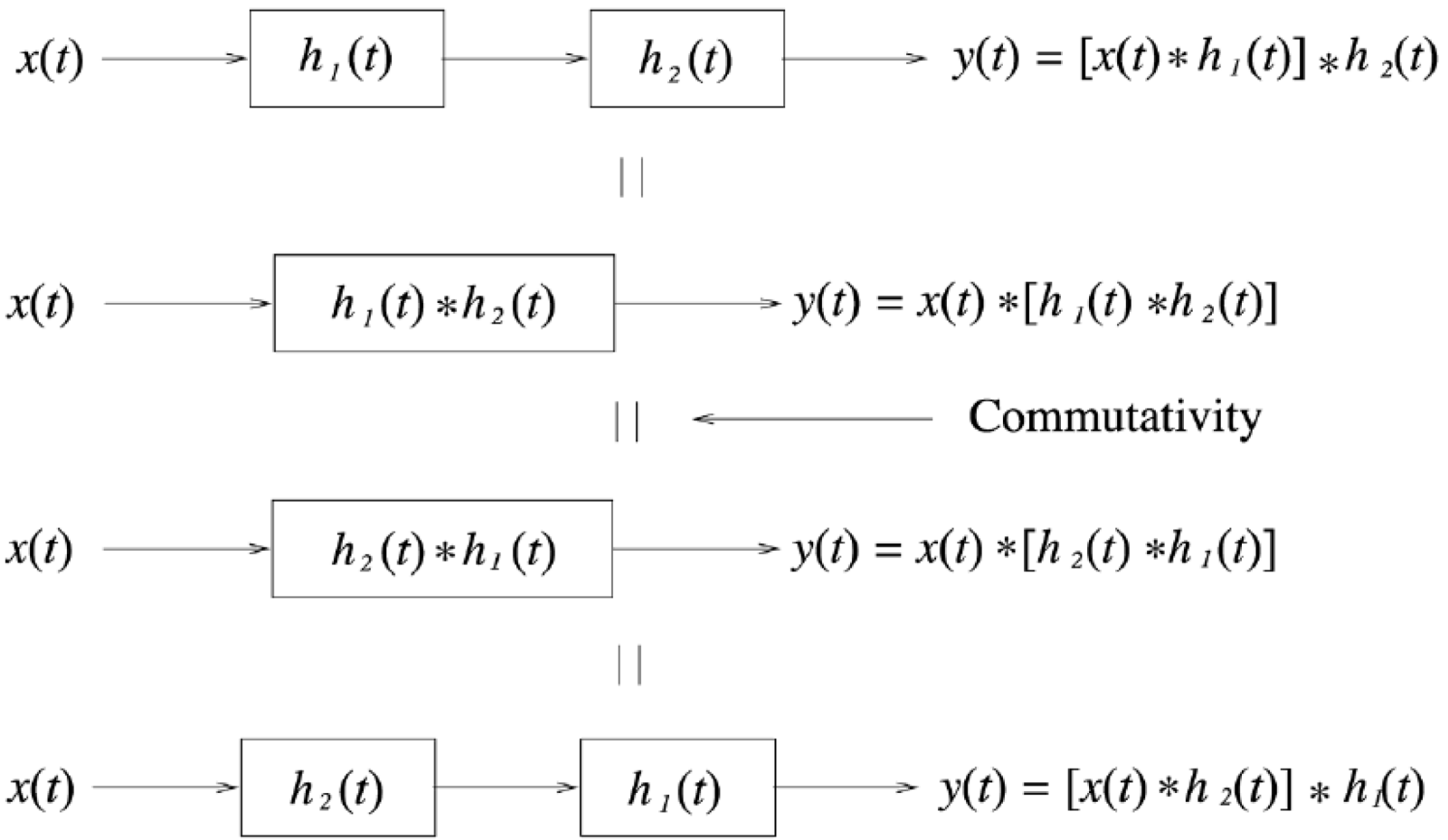
$$= e^{j\omega t}\underbrace{\int_{-\infty}^{\infty} e^{-j\omega\tau}h(\tau)d\tau}_{H(j\omega)}$$

Frequency response of the LTI system.

The frequency response of a CT LTI system is simply the Fourier transform of its impulse response

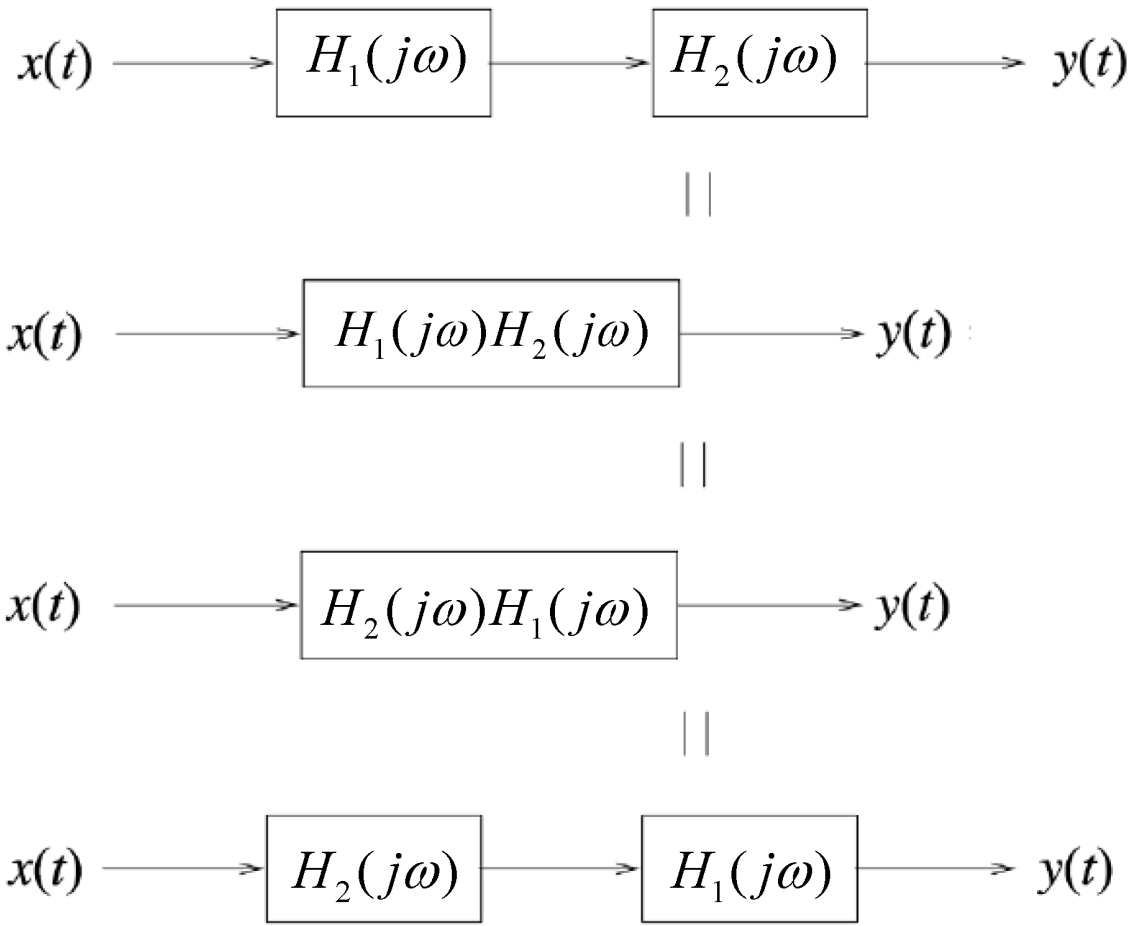
CT Fourier Transform

ASSOCIATIVITY



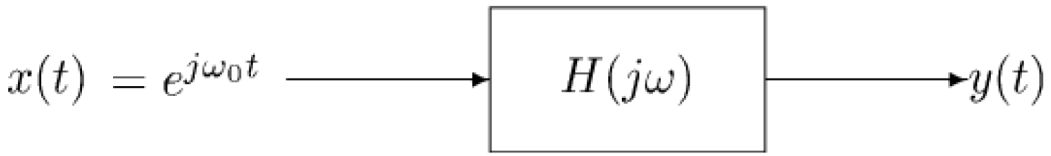
CT Fourier Transform

ASSOCIATIVITY



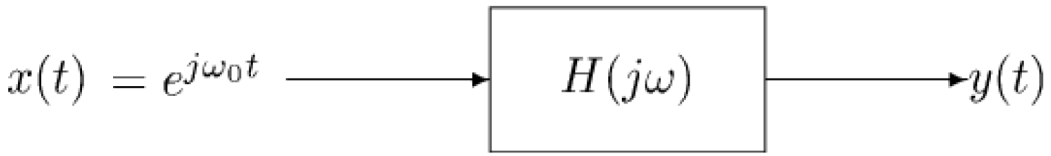
CT Fourier Transform

Ex.#1 **Determine** $y(t)$



CT Fourier Transform

Ex.#1 **Determine** $y(t)$



$$e^{j\omega_0 t} \longleftrightarrow 2\pi\delta(\omega - \omega_0)$$

$$Y(j\omega) = H(j\omega)X(j\omega) = H(j\omega)2\pi\delta(\omega - \omega_0) = 2\pi H(j\omega_0)\delta(\omega - \omega_0)$$

$$\Downarrow \text{ inverse FT}$$

$$y(t) = H(j\omega_0)e^{j\omega_0 t}$$

Ex.#2 Determine the frequency response.

A differentiator - **an LTI system**

$$y(t) = \frac{dx(t)}{dt}$$

CT Fourier Transform

Ex.#2 Determine the frequency response.

A differentiator **- an LTI system**

$$y(t) = \frac{dx(t)}{dt}$$

Differentiation property: $Y(j\omega) = j\omega X(j\omega)$

$$\Downarrow$$

$$H(j\omega) = j\omega$$

1) Amplifies high frequencies (enhances sharp edges)

2) $+\pi/2$ phase shift $(j = e^{j\pi/2})$

Larger at high ω_o — phase shift

$$\frac{d}{dt}\sin\omega_0 t = \omega_0\cos\omega_0 t = \omega_0\sin(\omega_0 t + \tfrac{\pi}{2})$$

$$\frac{d}{dt}\cos\omega_0 t = -\omega_0\sin\omega_0 t = \omega_0\cos(\omega_0 t + \tfrac{\pi}{2})$$

Assignments

- Reading Assignment: Ch. 4